UNIT – III

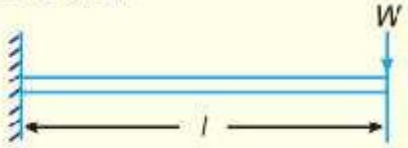
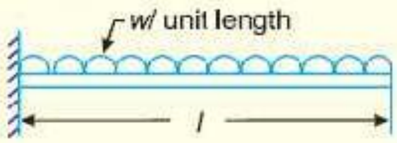
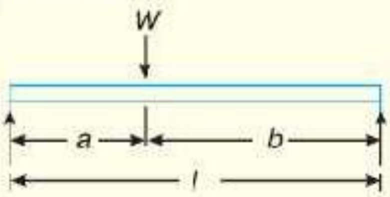
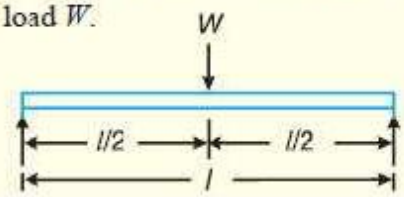
TRANSVERSE VIBRATION

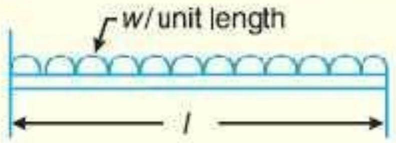
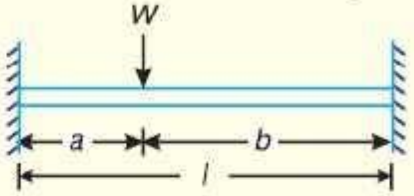
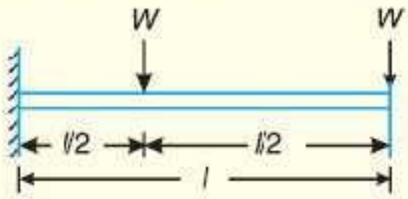
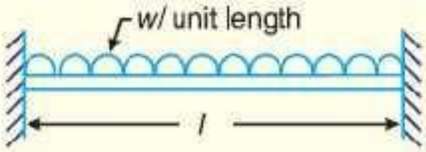
Time period, $t_p = 2\pi\sqrt{\frac{m}{s}}$

and natural frequency, $f_n = \frac{1}{t_p} = \frac{1}{2\pi}\sqrt{\frac{s}{m}} = \frac{1}{2\pi}\sqrt{\frac{g}{\delta}}$

If δ is the static deflection due to load W, then the natural frequency of the free transverse vibration is

$$f_n = \frac{1}{2\pi}\sqrt{\frac{g}{\delta}} = \frac{0.4985}{\sqrt{\delta}} \text{ Hz} \qquad \ldots \text{(Substituting, } g = 9.81 \text{ m/s}^2\text{)}$$

S.No.	*Type of beam*	*Deflection* (δ)
1.	Cantilever beam with a point load W at the free end.  	$\delta = \frac{Wl^3}{3EI}$ (at the free end)
2.	Cantilever beam with a uniformly distributed load of w per unit length.  	$\delta = \frac{wl^4}{8EI}$ (at the free end)
3.	Simply supported beam with an eccentric point load W.  	$\delta = \frac{Wa^2b^2}{3EIl}$ (at the point load)
4.	Simply supported beam with a central point load W.  	$\delta = \frac{Wl^3}{48EI}$ (at the centre)

S.No.	Type of beam	Deflection (δ)
5.	Simply supported beam with a uniformly distributed load of w per unit length. 	$\delta = \frac{5}{384} \times \frac{wl^4}{EI}$ (at the centre)
6.	Fixed beam with an eccentric point load W. 	$\delta = \frac{Wa^3b^3}{3EIl}$ (at the point load)
7.	Fixed beam with a central point load W. 	$\delta = \frac{Wl^3}{192EI}$ (at the centre)
8.	Fixed beam with a uniformly distributed load of w per unit length. 	$\delta = \frac{wl^4}{384EI}$ (at the centre)

PROBLEMS ON TRANSVERSE VIBRATIONS:

1. **A cantilever shaft 50 mm diameter and 300 mm long has a disc of mass 100 kg at its free end. The Young's modulus for the shaft material is 200 GN/m^2. Determine the frequency of longitudinal and transverse vibrations of the shaft.**

Solution. Given : d = 50 mm = 0.05 m ; l = 300 mm = 0.03 m ; m = 100 kg ; $E = 200$ GN/m^2 $= 200 \times 10^9$ N/m^2

We know that cross-sectional area of the shaft,

$$A = \frac{\pi}{4} \times d^2 = \frac{\pi}{4}(0.05)^2 = 1.96 \times 10^{-3}\,\text{m}^2$$

and moment of inertia of the shaft,

$$I = \frac{\pi}{64} \times d^4 = \frac{\pi}{64}(0.05)^4 = 0.3 \times 10^{-6}\,\text{m}^4$$

Frequency of longitudinal vibration

We know that static deflection of the shaft,

$$\delta = \frac{W.l}{A.E} = \frac{100 \times 9.81 \times 0.3}{1.96 \times 10^{-3} \times 200 \times 10^9} = 0.751 \times 10^{-6}\ \text{m}$$

$$\ldots (\because\ W = m.g)$$

$\therefore$ Frequency of longitudinal vibration,

$$f_n = \frac{0.4985}{\sqrt{\delta}} = \frac{0.4985}{\sqrt{0.751 \times 10^{-6}}} = 575\ \text{Hz}\ \ \textbf{Ans.}$$

Frequency of transverse vibration

We know that static deflection of the shaft,

$$\delta = \frac{W.l^3}{3\,E.I} = \frac{100 \times 9.81 \times (0.3)^3}{3 \times 200 \times 10^9 \times 0.3 \times 10^{-6}} = 0.147 \times 10^{-3}\ \text{m}$$

$\therefore$ Frequency of transverse vibration,

$$f_n = \frac{0.4985}{\sqrt{\delta}} = \frac{0.4985}{\sqrt{0.147 \times 10^{-3}}} = 41\ \text{Hz}\ \ \textbf{Ans.}$$

2. A shaft of length 0.75 m, supported freely at the ends, is carrying a body of mass 90 kg at 0.25 m from one end. Find the natural frequency of transverse vibration. Assume E = 200 GN/m^2 and shaft diameter = 50 mm.

Solution. Given : $l = 0.75$ m ; $m = 90$ kg ; $a = AC = 0.25$ m ; $E = 200$ GN/m^2 = 200×10^9 N/m^2; $d = 50$ mm = 0.05 m

The shaft is shown in Fig. 23.7.

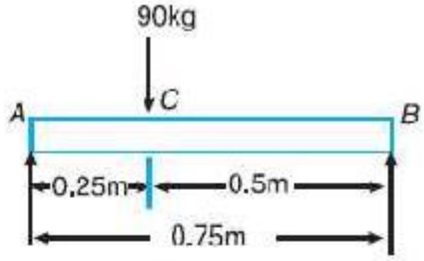


Fig.

We know that moment of inertia of the shaft,

$$I = \frac{\pi}{64} \times d^4 = \frac{\pi}{64}(0.05)^4 \text{ m}^4$$

$$= 0.307 \times 10^{-6} \text{ m}^4$$

and static deflection at the load point (*i.e.* at point C),

$$\delta = \frac{Wa^2b^2}{3EIl} = \frac{90 \times 9.81 (0.25)^2 (0.5)^2}{3 \times 200 \times 10^9 \times 0.307 \times 10^{-6} \times 0.75} = 0.1 \times 10^{-3} \text{ m}$$

. . . ($\because b = BC = 0.5$ m)

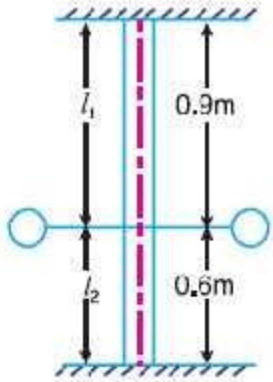
We know that natural frequency of transverse vibration,

$$f_n = \frac{0.4985}{\sqrt{\delta}} = \frac{0.4985}{\sqrt{0.1 \times 10^{-3}}} = 49.85 \text{ Hz} \quad \textbf{Ans.}$$

3. A flywheel is mounted on a vertical shaft as shown in Fig. 23.8. The both ends of the shaft are fixed and its diameter is 50 mm. The flywheel has a mass of 500 kg. Find the natural frequencies of longitudinal and transverse vibrations. Take E = 200 GN/m^2.

Solution. Given : $d = 50$ mm = 0.05 m ; $m = 500$ kg ; $E = 200$ GN/m^2 = 200×10^9 N/m^2

We know that cross-sectional area of shaft,



$$A = \frac{\pi}{4} \times d^2 = \frac{\pi}{4}(0.05)^2 = 1.96 \times 10^{-3} \text{ m}^2$$

and moment of inertia of shaft,

$$I = \frac{\pi}{64} \times d^4 = \frac{\pi}{64}(0.05)^4 = 0.307 \times 10^{-6} \text{ m}^4$$

Natural frequency of longitudinal vibration

Let m_1 = Mass of flywheel carried by the length l_1.

$\therefore$ $m - m_1$ = Mass of flywheel carried by length l_2.

We know that extension of length l_1

$$= \frac{W_1 . l_1}{A.E} = \frac{m_1 . g . l_1}{A.E} \quad \ldots (i)$$

Similarly, compression of length l_2

$$= \frac{(W - W_1)\, l_2}{A.E} = \frac{(m - m_1)\, g . l_2}{A.E} \quad \ldots (ii)$$

Since extension of length l_1 must be equal to compression of length l_2, therefore equating equations (*i*) and (*ii*),

$$m_1.l_1 = (m - m_1)l_2$$

$$m_1 \times 0.9 = (500 - m_1)0.6 = 300 - 0.6\,m_1 \quad \text{or} \quad m_1 = 200 \text{ kg}$$

$\therefore$ Extension of length l_1,

$$\delta = \frac{m_1.g.l_1}{A.E} = \frac{200 \times 9.81 \times 0.9}{1.96 \times 10^{-3} \times 200 \times 10^{9}} = 4.5 \times 10^{-6} \text{ m}$$

We know that natural frequency of longitudinal vibration,

$$f_n = \frac{0.4985}{\sqrt{\delta}} = \frac{0.4985}{\sqrt{4.5 \times 10^{-6}}} = 235 \text{ Hz} \quad \textbf{Ans.}$$

Natural frequency of transverse vibration

We know that the static deflection for a shaft fixed at both ends and carrying a point load is given by

$$\delta = \frac{Wa^3b^3}{3EIl^3} = \frac{500 \times 9.81 (0.9)^3 (0.6)^3}{3 \times 200 \times 10^9 \times 0.307 \times 10^{-6} (1.5)^3} = 1.24 \times 10^{-3} \text{ m}$$

. . . (Substituting $W = m.g$; $a = l_1$, and $b = l_2$)

We know that natural frequency of transverse vibration,

$$f_n = \frac{0.4985}{\sqrt{\delta}} = \frac{0.4985}{\sqrt{1.24 \times 10^{-3}}} = 14.24 \text{ Hz} \quad \textbf{Ans.}$$

<u>Natural Frequency of Free Transverse Vibrations For a Shaft Subjected to a Number of Point Loads</u>

<u>DUNKERLEY'S METHOD:</u>

The natural frequency of transverse vibration for a shaft carrying a number of point loads and uniformly distributed load is obtained from Dunkerley's empirical formula. According to this

$$\frac{1}{(f_n)^2} = \frac{1}{(f_{n1})^2} + \frac{1}{(f_{n2})^2} + \frac{1}{(f_{n3})^2} + \ldots + \frac{1}{(f_{ns})^2}$$

where f_n = Natural frequency of transverse vibration of the shaft carrying point loads and uniformly distributed load.

f_{n1}, f_{n2}, f_{n3}, etc. = Natural frequency of transverse vibration of each point load.

f_{ns} = Natural frequency of transverse vibration of the uniformly distributed load (or due to the mass of the shaft).

Now, consider a shaft *AB* loaded as shown in Fig.

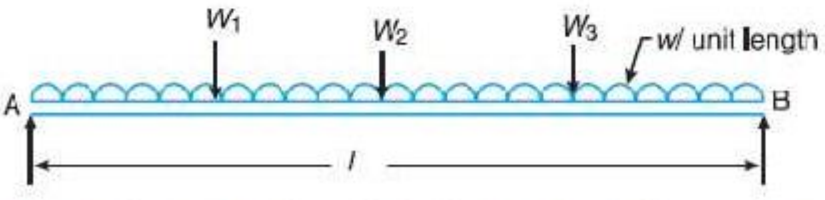


Fig. Shaft carrying a number of point loads and a uniformly distributed load.

Let $\delta_1, \delta_2, \delta_3$, etc. = Static deflection due to the load W_1, W_2, W_3 etc. when considered separately.

δ_S = Static deflection due to the uniformly distributed load or due to the mass of the shaft.

Therefore, according to Dunkerley's empirical formula, the natural frequency of the whole system,

$$\frac{1}{(f_n)^2} = \frac{1}{(f_{n1})^2} + \frac{1}{(f_{n2})^2} + \frac{1}{(f_{n3})^3} + \dots + \frac{1}{(f_{ns})^2}$$

$$= \frac{\delta_1}{(0.4985)^2} + \frac{\delta_2}{(0.4985)^2} + \frac{\delta_3}{(0.4985)^2} + \dots + \frac{\delta_S}{(0.5615)^2}$$

$$= \frac{1}{(0.4985)^2}\left[\delta_1 + \delta_2 + \delta_3 + \dots + \frac{\delta_S}{1.27}\right]$$

When there is no uniformly distributed load or mass of the shaft is negligible, then $\delta_S = 0$.

$$f_n = \frac{0.4985}{\sqrt{\delta_1 + \delta_2 + \delta_3 + \dots}} \text{ Hz}$$

PROBLEM ON DUNKERLEY'S EQUATION:

1. **A shaft 50 mm diameter and 3 metres long is simply supported at the ends and carries three loads of 1000 N, 1500 N and 750 N at 1 m, 2 m and 2.5 m from the left support. The Young's modulus for shaft material is 200 GN/m^2. Find the frequency of transverse vibration.**

Solution. Given : d = 50 mm = 0.05 m ; l = 3 m, W_1 = 1000 N ; W_2 = 1500 N ; W_3 = 750 N; E = 200 GN/m^2 = 200 × 10^9 N/m^2

The shaft carrying the loads is shown in Fig.

We know that moment of inertia of the shaft,

$$I = \frac{\pi}{64} \times d^4 = \frac{\pi}{64}(0.05)^4 = 0.307 \times 10^{-6} \text{ m}^4$$

and the static deflection due to a point load W,

$$\delta = \frac{Wa^2b^2}{3EIl}$$

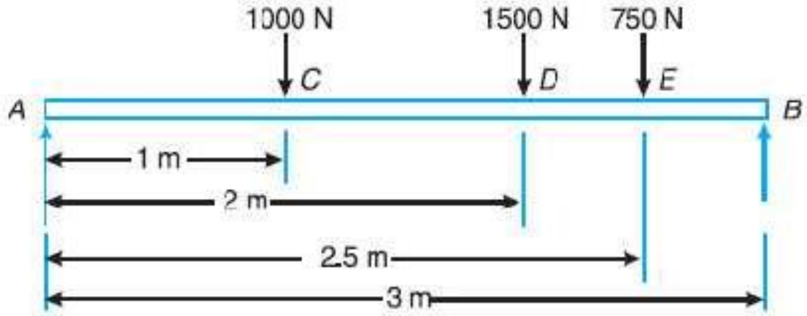


Fig.

∴ Static deflection due to a load of 1000 N,

$$\delta_1 = \frac{1000 \times 1^2 \times 2^2}{3 \times 200 \times 10^9 \times 0.307 \times 10^{-6} \times 3} = 7.24 \times 10^{-3} \text{ m}$$

... (Here a = 1 m, and b = 2 m)

Similarly, static deflection due to a load of 1500 N,

$$\delta_2 = \frac{1500 \times 2^2 \times 1^2}{3 \times 200 \times 10^9 \times 0.307 \times 10^{-6} \times 3} = 10.86 \times 10^{-3} \text{ m}$$

. . . (Here $a = 2$ m, and $b = 1$ m)

and static deflection due to a load of 750 N,

$$\delta_3 = \frac{750\,(2.5)^2\,(0.5)^2}{3 \times 200 \times 10^9 \times 0.307 \times 10^{-6} \times 3} = 2.12 \times 10^{-3} \text{ m}$$

. . . (Here $a = 2.5$ m, and $b = 0.5$ m)

We know that frequency of transverse vibration,

$$f_n = \frac{0.4985}{\sqrt{\delta_1 + \delta_2 + \delta_3}} = \frac{0.4985}{\sqrt{7.24 \times 10^{-3} + 10.86 \times 10^{-3} + 2.12 \times 10^{-3}}}$$

$$= \frac{0.4985}{0.1422} = 3.5 \text{ Hz} \quad \textbf{Ans.}$$

CRITICAL OR WHIRLING SPEED OF A SHAFT

Critical or whirling speed,

$$\omega_c = \omega_n = \sqrt{\frac{s}{m}} = \sqrt{\frac{g}{\delta}} \text{ Hz} \qquad \ldots \left(\because\ \delta = \frac{m.g}{s}\right)$$

If N_c is the critical or whirling speed in r.p.s., then

$$2\pi N_c = \sqrt{\frac{g}{\delta}} \quad \text{or} \quad N_c = \frac{1}{2\pi}\sqrt{\frac{g}{\delta}} = \frac{0.4985}{\sqrt{\delta}} \text{ r.p.s.}$$

where δ = Static deflection of the shaft in metres.

PROBLEMS ON CRITICAL SPEED OF A SHAFT:

1. **Calculate the whirling speed of a shaft 20 mm diameters and 0.6 m long carrying a mass of 1 kg at its mid-point. The density of the shaft material is 40 Mg/m^3, and Young's modulus is 200 GN/m^2. Assume the shaft to be freely supported.**

Solution. Given : d = 20 mm = 0.02 m ; l = 0.6 m ; m_1 = 1 kg ; ρ = 40 Mg/m^3 = 40 × 10^6 g/m^3 = 40 × 10^3 kg/m^3 ; E = 200 GN/m^2 = 200 × 10^9 N/m^2

The shaft is shown in Fig. .

We know that moment of inertia of the shaft,

$$I = \frac{\pi}{64} \times d^4 = \frac{\pi}{64}(0.02)^4 \text{ m}^4$$

$$= 7.855 \times 10^{-9} \text{ m}^4$$

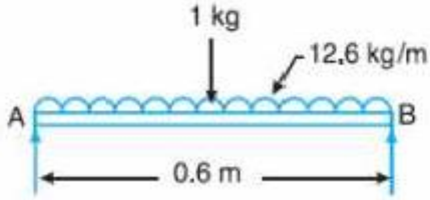


Fig.

Since the density of shaft material is 40 × 10^3 kg/m^3, therefore mass of the shaft per metre length,

$$m_S = \text{Area} \times \text{length} \times \text{density} = \frac{\pi}{4}(0.02)^2 \times 1 \times 40 \times 10^3 = 12.6 \text{ kg/m}$$

We know that static deflection due to 1 kg of mass at the centre,

$$\delta = \frac{Wl^3}{48\,EI} = \frac{1 \times 9.81(0.6)^3}{48 \times 200 \times 10^9 \times 7.855 \times 10^{-9}} = 28 \times 10^{-6} \text{ m}$$

and static deflection due to mass of the shaft,

$$\delta_S = \frac{5wl^4}{384\,EI} = \frac{5 \times 12.6 \times 9.81(0.6)^4}{384 \times 200 \times 10^9 \times 7.855 \times 10^{-9}} = 0.133 \times 10^{-3} \text{ m}$$

$\therefore$ Frequency of transverse vibration,

$$f_n = \frac{0.4985}{\sqrt{\delta + \frac{\delta_S}{1.27}}} + \frac{0.4985}{\sqrt{28 \times 10^{-6} + \frac{0.133 \times 10^{-3}}{1.27}}}$$

$$= \frac{0.4985}{11.52 \times 10^{-3}} = 43.3 \text{ Hz}$$

Let N_c = Whirling speed of a shaft.

We know that whirling speed of a shaft in r.p.s. is equal to the frequency of transverse vibration in Hz , therefore

$$N_c = 43.3 \text{ r.p.s.} = 43.3 \times 60 = 2598 \text{ r.p.m.}$$ **Ans.**

2. **A shaft 1.5 m long, supported in flexible bearings at the ends carries two wheels each of 50 kg mass. One wheel is situated at the centre of the shaft and the other at a distance of 375 mm from the centre towards left. The shaft is hollow of external diameter 75 mm and internal diameter 40 mm. The density of the shaft material is 7700 kg/m^3 and its modulus of elasticity is 200 GN/m^2. Find the lowest whirling speed of the shaft, taking into account the mass of the shaft.**

Solution. l = 1.5 m ; $m_1 = m_2$ = 50 kg ; d_1 = 75 mm = 0.075 m ; d_2 = 40 mm = 0.04 m ; ρ = 7700 kg/m^3 ; E = 200 GN/m^2 = 200 × 10^9 N/m^2

The shaft is shown in Fig.

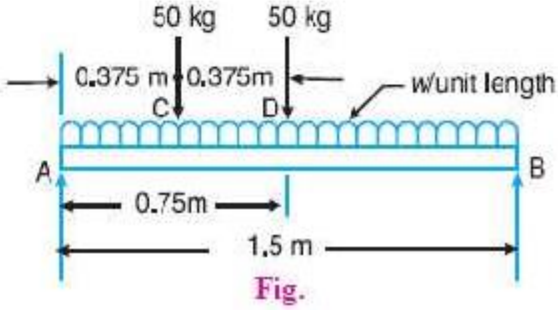


Fig.

We know that moment of inertia of the shaft,

$$I = \frac{\pi}{64}\left[(d_1)^4 - (d_2)^4\right] = \frac{\pi}{64}\left[(0.075)^4 - (0.04)^4\right] = 1.4\times10^{-6}\,\text{m}^4$$

Since the density of shaft material is 7700 kg/m^3, therefore mass of the shaft per metre length,

$$m_S = \text{Area} \times \text{length} \times \text{density}$$

$$= \frac{\pi}{4}\left[(0.075)^2 - (0.04)^2\right]1\times7700 = 24.34\text{ kg/m}$$

We know that the static deflection due to a load W

$$= \frac{Wa^2b^2}{3EIl} = \frac{m.ga^2b^2}{3EIl}$$

∴ Static deflection due to a mass of 50 kg at C,

$$\delta_1 = \frac{m_1ga^2b^2}{3EIl} = \frac{50\times9.81(0.375)^2(1.125)^2}{3\times200\times10^9\times1.4\times10^{-6}\times1.5} = 70\times10^{-6}\text{ m}$$

. . . (Here a = 0.375 m, and b = 1.125 m)

Similarly, static deflection due to a mass of 50 kg at D

$$\delta_2 = \frac{m_1ga^2b^2}{3EIl} = \frac{50\times9.81(0.75)^2(0.75)^2}{3\times200\times10^9\times1.4\times10^{-6}\times1.5} = 123\times10^{-6}\text{ m}$$

. . . (Here $a = b$ = 0.75 m)

We know that static deflection due to uniformly distributed load or mass of the shaft,

$$\delta_S = \frac{5}{384}\times\frac{wl^4}{EI} = \frac{5}{384}\times\frac{24.34\times9.81(1.5)^4}{200\times10^9\times1.4\times10^{-6}} = 56\times10^{-6}\text{ m}$$

. . . (Substituting, $w = m_S \times g$)

We know that frequency of transverse vibration,

$$f_n = \frac{0.4985}{\sqrt{\delta_1 + \delta_2 + \dfrac{\delta_S}{1.27}}} = \frac{0.4985}{\sqrt{70\times10^{-6} + 123\times10^{-6} + \dfrac{56\times10^{-6}}{1.27}}}\text{ Hz}$$

$$= 32.4\text{ Hz}$$

Since the whirling speed of shaft (N_c) in r.p.s. is equal to the frequency of transverse vibration in Hz, therefore

$$N_c = 32.4\text{ r.p.s.} = 32.4\times60 = 1944\text{ r.p.m.}\quad \textbf{Ans.}$$

3. **A vertical shaft of 5 mm diameter is 200 mm long and is supported in long bearings at its ends. A disc of mass 50 kg is attached to the centre of the shaft. Neglecting any increase in stiffness due to the attachment of the disc to the shaft, find the critical speed of rotation and the maximum bending stress when the shaft is rotating at 75% of the critical speed. The centre of the disc is 0.25 mm from the geometric axis of the shaft. E = 200 GN/m^2.**

Solution. Given : $d = 5$ mm $= 0.005$ m ; $l = 200$ mm $= 0.2$ m ; $m = 50$ kg ; $e = 0.25$ mm $= 0.25 \times 10^{-3}$ m ; $E = 200$ GN/m^2 $= 200 \times 10^9$ N/m^2

Critical speed of rotation

We know that moment of inertia of the shaft,

$$I = \frac{\pi}{64} \times d^4 = \frac{\pi}{64}(0.005)^4 = 30.7 \times 10^{-12} \text{m}^4$$

Since the shaft is supported in long bearings, it is assumed to be fixed at both ends. We know that the static deflection at the centre of the shaft due to a mass of 50 kg,

$$\delta = \frac{Wl^3}{192EI} = \frac{50 \times 9.81(0.2)^3}{192 \times 200 \times 10^9 \times 30.7 \times 10^{-12}} = 3.33 \times 10^{-3} \text{m}$$

. . . ($\because W = m.g$)

We know that critical speed of rotation (or natural frequency of transverse vibrations),

$$N_c = \frac{0.4985}{\sqrt{3.33 \times 10^{-3}}} = 8.64 \text{ r.p.s.}$$ **Ans.**

Maximum bending stress

Let σ = Maximum bending stress in N/m^2, and

N = Speed of the shaft = 75% of critical speed = $0.75\,N_c$ . . . (Given)

When the shaft starts rotating, the additional dynamic load (W_1) to which the shaft is subjected, may be obtained by using the bending equation,

$$\frac{M}{I} = \frac{\sigma}{y_1} \quad \text{or} \quad M = \frac{\sigma . I}{y_1}$$

We know that for a shaft fixed at both ends and carrying a point load (W_1) at the centre, the maximum bending moment

$$M = \frac{W_1 . l}{8}$$

$$\therefore \qquad \frac{W_1 . l}{8} = \frac{\sigma \cdot I}{d/2} \qquad \ldots (\because\ y_1 = d/2)$$

and
$$W_1 = \frac{\sigma I}{d/2} \times \frac{8}{l} = \frac{\sigma \times 30.7 \times 10^{-12}}{0.005/2} \times \frac{8}{0.2} = 0.49 \times 10^{-6}\, \sigma\, \text{N}$$

$\therefore$ Additional deflection due to load W_1,

$$y = \frac{W_1}{W} \times \delta = \frac{0.49 \times 10^{-6} \sigma}{50 \times 9.81} \times 3.33 \times 10^{-3} = 3.327 \times 10^{-12} \sigma$$

We know that

$$y = \frac{\pm e}{\left(\frac{\omega_c}{\omega}\right)^2 - 1} = \frac{\pm e}{\left(\frac{N_c}{N}\right)^2 - 1} \qquad \ldots \text{(Substituting } \omega_c = N_c \text{ and } \omega = N\text{)}$$

$$3.327 \times 10^{-12} \sigma = \frac{\pm 0.25 \times 10^{-3}}{\left(\frac{N_c}{0.75 N_c}\right)^2 - 1} = \pm 0.32 \times 10^{-3}$$

$$\sigma = 0.32 \times 10^{-3} / 3.327 \times 10^{-12} = 0.0962 \times 10^9 \text{ N/m}^2 \quad \ldots \text{(Taking + ve sign)}$$

$$= 96.2 \times 10^6 \text{ N/m}^2 = 96.2 \text{ MN/m}^2 \textbf{ Ans.}$$

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4. A vertical steel shaft 15 mm diameter is held in long bearings 1 metre apart and carries at its middle a disc of mass 15 kg. The eccentricity of the centre of gravity of the disc from the centre of the rotor is 0.30 mm. The modulus of elasticity for the shaft material is 200 GN/m^2 and the permissible stress is 70 MN/m^2. Determine: 1. the critical speed of the shaft and 2. The range of speed over which it is unsafe to run the shaft. Neglect the mass of the shaft. [For a shaft with fixed end carrying a concentrated load (W) at the centre assumes $\delta = \frac{Wl^3}{192\,EI}$ and $= \frac{Wl}{8}$, where δ and M are maximum deflection and bending moment respectively].

Solution. Given : d = 15 mm = 0.015 m ; l = 1 m ; m = 15 kg ; e = 0.3 mm = 0.3 × 10^{-3} m ; E = 200 GN/m^2 = 200 × 10^9 N/m^2 ; σ = 70 MN/m^2 = 70 × 10^6 N/m^2

We know that moment of inertia of the shaft,

$$I = \frac{\pi}{64} \times d^4 = \frac{\pi}{64}(0.015)^4 = 2.5 \times 10^{-9} \text{ m}^4$$

1. Critical speed of the shaft

Since the shaft is held in long bearings, therefore it is assumed to be fixed at both ends. We know that the static deflection at the centre of shaft,

$$\delta = \frac{Wl^3}{192\,EI} = \frac{15 \times 9.81 \times 1^3}{192 \times 200 \times 10^9 \times 2.5 \times 10^{-9}} = 1.5 \times 10^{-3} \text{m} \qquad \ldots (\because\ W = m.g)$$

$\therefore$ Natural frequency of transverse vibrations,

$$f_n = \frac{0.4985}{\sqrt{\delta}} = \frac{0.4985}{\sqrt{1.5 \times 10^{-3}}} = 12.88 \text{ Hz}$$

We know that the critical speed of the shaft in r.p.s. is equal to the natural frequency of transverse vibrations in Hz.

$\therefore$ Critical speed of the shaft,

$$N_c = 12.88 \text{ r.p.s.} = 12.88 \times 60 = 772.8 \text{ r.p.m. } \textbf{Ans.}$$

2. ***Range of speed***

Let N_1 and N_2 = Minimum and maximum speed respectively.

When the shaft starts rotating, the additional dynamic load $(W_1 = m_1.g)$ to which the shaft is subjected may be obtained from the relation

$$\frac{M}{I} = \frac{\sigma}{y_1} \quad \text{or} \quad M = \frac{\sigma . I}{y_1}$$

Since $M = \frac{W_1 . l}{8} = \frac{m_1 . g . l}{8}$, and $y_1 = \frac{d}{2}$, therefore

$$\frac{m_1 . g . l}{8} = \frac{\sigma I}{d/2}$$

or

$$m_1 = \frac{8 \times 2 \times \sigma \times I}{d . g . l} = \frac{8 \times 2 \times 70 \times 10^6 \times 2.5 \times 10^{-9}}{0.015 \times 9.81 \times 1} = 19 \text{ kg}$$

$\therefore$ Additional deflection due to load $W_1 = m_1 g$,

$$y = \frac{W_1}{W} \times \delta = \frac{m_1}{m} \times \delta = \frac{19}{15} \times 1.5 \times 10^{-3} = 1.9 \times 10^{-3} \text{ m}$$

We know that,

$$y = \frac{\pm e}{\left(\frac{\omega_c}{\omega}\right)^2 - 1} \quad \text{or} \quad \pm \frac{y}{e} = \frac{1}{\left(\frac{N_c}{N}\right)^2 - 1}$$

(Substituting $\omega_c = N_c$, and $\omega = N$)

$$\therefore \quad \pm \frac{1.9 \times 10^{-3}}{0.3 \times 10^{-3}} = \frac{1}{\left(\frac{N_c}{N}\right)^2 - 1} \quad \text{or} \quad \left(\frac{N_c}{N}\right)^2 - 1 = \pm \frac{0.3}{1.9} = \pm 0.16$$

$$\left(\frac{N_c}{N}\right)^2 = 1 \pm 0.16 = 1.16 \quad \text{or} \quad 0.84$$

... (Taking first plus sign and then negative sign)

or

$$N = \frac{N_c}{\sqrt{1.16}} \quad \text{or} \quad \frac{N_c}{\sqrt{0.84}}$$

$$\therefore \quad N_1 = \frac{N_c}{\sqrt{1.16}} = \frac{772.8}{\sqrt{1.16}} = 718 \text{ r.p.m.}$$

and

$$N_2 = \frac{N_c}{\sqrt{0.84}} = \frac{772.8}{\sqrt{0.84}} = 843 \text{ r.p.m.}$$

Hence the range of speed is from 718 r.p.m. to 843 r.p.m. **Ans.**

TORSIONAL VIBRATION

$$\omega = \sqrt{\frac{q}{I}}$$

Time period, $t_p = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{I}{q}}$

natural frequency , $f_n = \frac{1}{t_p} = \frac{1}{2\pi}\sqrt{\frac{q}{I}}$

The value of the torsional stiffness q may be obtained from the torsion equation,

$$\frac{T}{J} = \frac{C.\theta}{l} \quad \text{or} \quad \frac{T}{\theta} = \frac{C.J}{l}$$

$$\therefore \quad q = \frac{C.J}{l} \qquad \left(\because \frac{T}{\theta} = q\right)$$

where C = Modulus of rigidity for the shaft material.

J = Polar moment of inertia of the shaft cross section,

$= \frac{\pi}{32} d^4$; d is the diameter of the shaft, and

l = Length of the shaft.

PROBLEMS:

1. **A shaft of 100 mm diameter and 1 metre long has one of its end fixed and the other end carries a disc of mass 500 kg at a radius of gyration of 450 mm. The modulus of rigidity for the shaft material is 80 GN/m^2. Determine the frequency of torsional vibrations.**

Solution. Given : d = 100 mm = 0.1 m ; l = 1 m ; m = 500 kg ; k = 450 mm = 0.45 m ; C = 80 GN/m^2 = 80×10^9 N/m^2

We know that polar moment of inertia of the shaft,

$$J = \frac{\pi}{32} \times d^4 = \frac{\pi}{32}(0.1)^4 = 9.82 \times 10^{-6} \text{ m}^4$$

$\therefore$ Torsional stiffness of the shaft,

$$q = \frac{C.J}{l} = \frac{80 \times 10^9 \times 9.82 \times 10^{-6}}{1} = 785.6 \times 10^3 \text{ N-m}$$

We know that mass moment of inertia of the shaft,

$$I = mk^2 = 500(0.45)^2 = 101.25 \text{ kg-m}^2$$

$\therefore$ Frequency of torsional vibrations,

$$f_n = \frac{1}{2\pi}\sqrt{\frac{q}{I}} = \frac{1}{2\pi}\sqrt{\frac{785.6 \times 10^3}{101.25}} = \frac{88.1}{2\pi} = 14 \text{ Hz}$$ **Ans.**

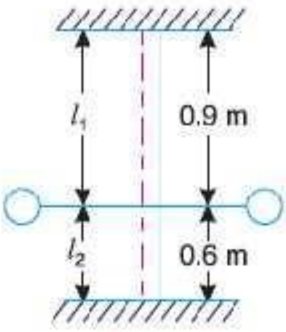
2. A flywheel is mounted on a vertical shaft as shown in Fig. The both ends of a shaft are fixed and its diameter is 50 mm. The flywheel has a mass of 500 kg and its radius of gyration is 0.5 m. Find the natural frequency of torsional vibrations, if the modulus of rigidity for the shaft material is 80 GN/m^2.

Solution. Given : $d = 50$ mm $= 0.05$ m ; $m = 500$ kg ; $k = 0.5$m; $G = 80$ GN/m^2 $= 84 \times 10^9$ N/m^2

We know that polar moment of inertia of the shaft,

$$J = \frac{\pi}{32} \times d^4 = \frac{\pi}{32}(0.05)^4 \text{ m}^4$$

$$= 0.6 \times 10^{-6} \text{ m}^4$$



Fig

∴ Torsional stiffness of the shaft for length l_1,

$$q_1 = \frac{C.J}{l_1} = \frac{84 \times 10^9 \times 0.6 \times 10^{-6}}{0.9}$$

$$= 56 \times 10^3 \text{ N-m}$$

Similarly torsional stiffness of the shaft for length l_2,

$$q_2 = \frac{C.J}{l_2} = \frac{84 \times 10^9 \times 0.6 \times 10^{-6}}{0.6} = 84 \times 10^3 \text{ N-m}$$

∴ Total torsional stiffness of the shaft,

$$q = q_1 + q_2 = 56 \times 10^3 + 84 \times 10^3 = 140 \times 10^3 \text{ N-m}$$

We know that mass moment of inertia of the flywheel,

$$I = m.k^2 = 500(0.5)^2 = 125 \text{ kg-m}^2$$

∴ Natural frequency of torsional vibration,

$$f_n = \frac{1}{2\pi}\sqrt{\frac{q}{I}} = \frac{1}{2\pi}\sqrt{\frac{140 \times 10^3}{125}} = \frac{33.5}{2\pi} = 5.32 \text{ Hz}$$ **Ans.**

FREE TORSIONAL VIBRATIONS OF A SINGLE ROTOR SYSTEM

$$f_n \quad \frac{1}{2}\sqrt{\frac{q}{I}} \quad \frac{1}{2}\sqrt{\frac{C.J}{l.I}}$$

$$\ldots \quad q \quad \frac{C.J}{l}$$

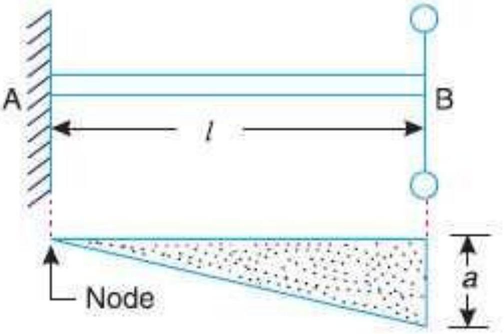


Fig Free torsional vibrations of a single rotor system.

where C = Modulus of rigidity for shaft material,

J = Polar moment of inertia of shaft

$$-\frac{}{32} \; d^4$$

d = Diameter of shaft,

l – Length of shaft,

m = Mass of rotor,

k = Radius of gyration of rotor, and

I = Mass moment of inertia of rotor = $m.k^2$

FREE TORSIONAL VIBRATIONS OF A TWO ROTOR SYSTEM

Let l = Length of shaft,

l_A = Length of part NP *i.e.* distance of node from rotor A,

l_B – Length of part NQ, *i.e.* distance of node from rotor B,

I_A = Mass moment of inertia of rotor A,

I_B = Mass moment of inertia of rotor B,

d – Diameter of shaft,

J = Polar moment of inertia of shaft, and

C = Modulus of rigidity for shaft material.

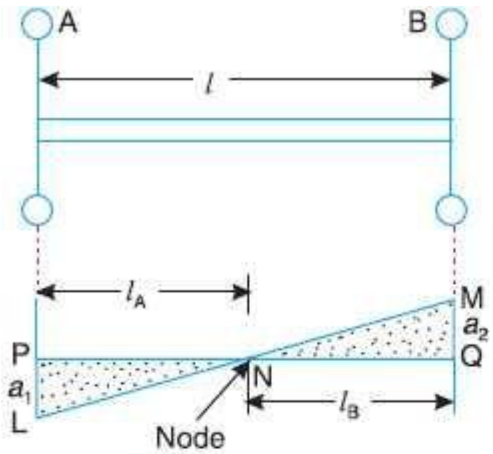


Fig Free torsional vibrations of a two rotor system.

Natural frequency of torsional vibration for rotor A,

$$f_{nA} \quad \frac{1}{2}\sqrt{\frac{C.J}{l_A.I_A}}$$

and natural frequency of torsional vibration for rotor B,

$$f_{nB} \quad \frac{1}{2}\sqrt{\frac{C.J}{l_B.I_B}}$$

Since $f_{nA} = f_{nB}$, therefore

$$\frac{1}{2}\sqrt{\frac{C.J}{l_A I_A}} \quad \frac{1}{2}\sqrt{\frac{C.J}{l_B.I_B}} \quad \text{or} \quad l_A \quad I_A = l_B \cdot I_B$$

$$l_A \quad \frac{l_B.I_B}{I_A}$$

FREE TORSIONAL VIBRATIONS OF A THREE ROTOR SYSTEM

Let l_1 = Distance between rotors A and B,
l_2 = Distance between rotors B and C,
l_A = Distance of node N_1 from rotor A,
l_C = Distance of node N_2 from rotor C,
I_A = Mass moment of inertia of rotor A,
I_B = Mass moment of inertia of rotor B,
I_C = Mass moment of inertia of rotor C,
d = Diameter of shaft,
J = Polar moment of inertia of shaft, and
C = Modulus of rigidity for shaft material

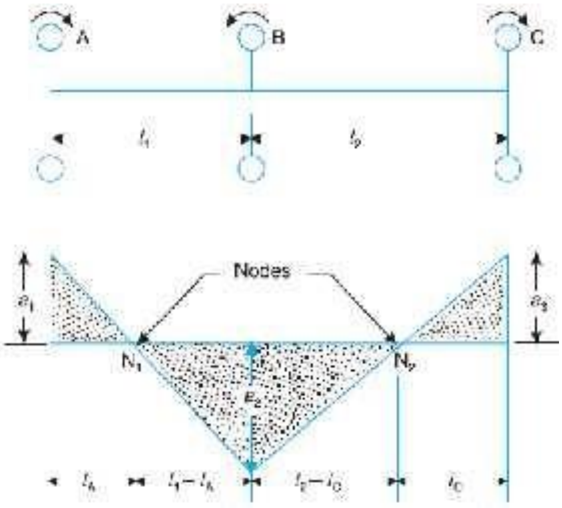


Fig. Free torsional vibrations of a three rotor system.

Natural frequency of torsional vibrations for rotor A,

$$f_{nA} \quad \frac{1}{2}\sqrt{\frac{C.J}{l_A.I_A}}$$

Natural frequency of torsional vibrations for rotor B,

$$f_{nB} \quad \frac{1}{2}\sqrt{\frac{C.J}{I_B}\quad\frac{1}{l_1 \quad l_A}\quad\frac{1}{l_2 \quad l_C}}$$

and natural frequency of torsional vibrations for rotor C,

$$f_{nC} \quad \frac{1}{2}\sqrt{\frac{C.J}{l_C.I_C}}$$

Since $f_{nA} = f_{nB} = f_{nC}$, therefore equating equations (*i*) and (*iii*)

$$\frac{1}{2}\sqrt{\frac{C.J}{l_A.I_A}} \quad \frac{1}{2}\sqrt{\frac{C.J}{l_C.I_C}} \qquad \text{or} \qquad l_A.I_A = l_C$$

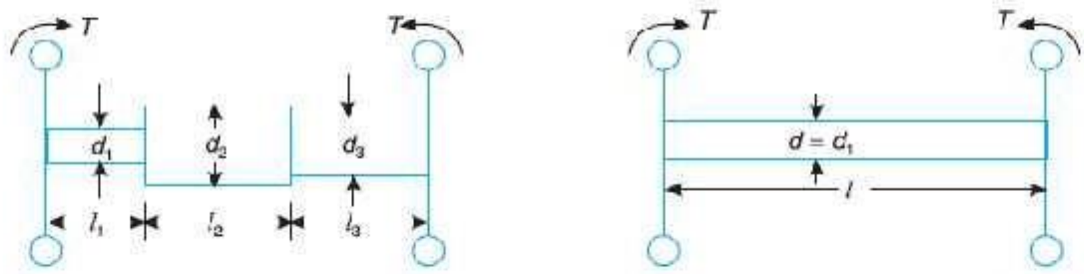
$$l_A \quad \frac{l_C.I_C}{I_A}$$

Now equating equations (*ii*) and (*iii*),

$$\frac{1}{2}\sqrt{\frac{C.J}{I_B}\quad\frac{1}{l_1 \quad l_A}\quad\frac{1}{l_2 \quad l_C}} \quad \frac{1}{2}\sqrt{\frac{C.J}{l_C.I_C}}$$

TORSIONALLY EQUIVALENT SHAFT

Let d_1, d_2 and d_3 = Diameters for the lengths l_1, l_2 and l_3 respectively,
$_1$, $_2$ and $_3$ = Angle of twist for the lengths l_1, l_2 and l_3 respectively,
= Total angle of twist, and
J_1, J_2 and J_3 = Polar moment of inertia for the shafts of diameters d_1, d_2 and d_3 respectively.



(*a*) Shaft of varying diameters. (*b*) Torsionally equivalent shaft.

1 2 3

or $$\frac{T.l}{C.J} \quad \frac{T.l_1}{C.J_1} \quad \frac{T.l_2}{C.J_2} \quad \frac{T.l_3}{C.J_3}$$

$$\frac{l}{J} \quad \frac{l_1}{J_1} \quad \frac{l_2}{J_2} \quad \frac{l_3}{J_3}$$

$$\frac{l}{\frac{}{32} d^4} \quad \frac{l_1}{\frac{}{32}(d_1)^4} \quad \frac{l_2}{\frac{}{32}(d_2)^4} \quad \frac{l_3}{\frac{}{32}(d_3)^4}$$

$$\frac{l}{d^4} \quad \frac{l_1}{(d_1)^4} \quad \frac{l_2}{(d_2)^4} \quad \frac{l_3}{(d_3)^4}$$

In actual calculations, it is assumed that the diameter d of the equivalent shaft is equal to one of the diameter of the actual shaft. Let us assume that $d - d_1$.

$$\frac{l}{(d_1)^4} \quad \frac{l_1}{(d_1)^4} \quad \frac{l_2}{(d_2)^4} \quad \frac{l_3}{(d_3)^4}$$

or $$l \quad l_1 \quad l_2 \quad \frac{d_1}{d_2}^4 \quad l_3 \quad \frac{d_1}{d_3}^4$$

This expression gives the length l of an equivalent shaft.

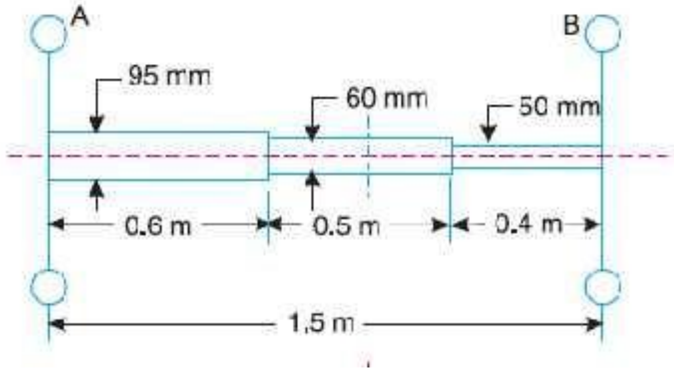
PROBLEMS:

Example 24.3. *A steel shaft 1.5 m long is 95 mm in diameter for the first 0.6 m of its length, 60 mm in diameter for the next 0.5 m of the length and 50 mm in diameter for the remaining 0.4 m of its length. The shaft carries two flywheels at two ends, the first having a mass of 900 kg and 0.85 m radius of gyration located at the 95 mm diameter end and the second having a mass of 700 kg and 0.55 m radius of gyration located at the other end. Determine the location of the node and the natural frequency of free torsional vibration of the system. The modulus of rigidity of shaft material may be taken as 80 GN/m².*

Solution. Given : $L - 1.5$ m ; $d_1 - 95$ mm $- 0.095$ m ; $l_1 - 0.6$ m ; $d_2 - 60$ mm $- 0.06$ m ; $l_2 = 0.5$ m ; $d_3 = 50$ mm $= 0.05$ m ; $l_3 = 0.4$ m ; $m_A = 900$ kg ; $k_A = 0.85$ m ; $m_B = 700$ kg ; $k_B = 0.55$ m ; $C = 80$ GN/m^2 = 80×10^9 N/m^2

The actual shaft is shown in Fig. 24.9 (*a*). First of all, let us find the length of the equivalent shaft, assuming its diameter as $d_1 = 95$ mm as shown in Fig 24.9 (*b*).

(*a*)



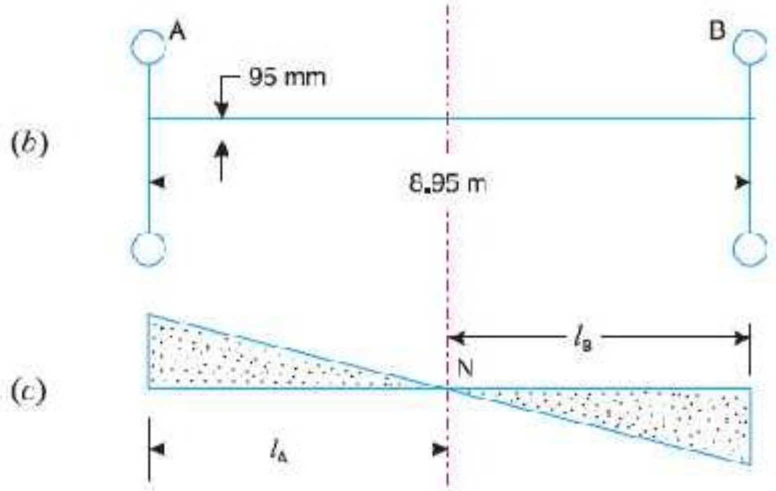


Fig. 24.9

We know that length of the equivalent shaft,

$$l = l_1 + l_2\left(\frac{d_1}{d_2}\right)^4 + l_3\left(\frac{d_1}{d_3}\right)^4 = 0.6 + 0.5\left(\frac{0.095}{0.06}\right)^4 + 0.4\left(\frac{0.095}{0.05}\right)^4$$
$$= 0.6 + 3.14 + 5.21 = 8.95 \text{ m}$$

Location of the node

Suppose the node of the equivalent shaft lies at N as shown in Fig. 24.9 (c).

Let l_A = Distance of the node from flywheel A, and

l_B = Distance of the node from flywheel B.

We know that mass moment of inertia of flywheel A,

$$I_A = m_A (k_A)^2 = 900(0.85)^2 = 650 \text{ kg-m}^2$$

and mass moment of inertia of flywheel B,

$$I_B = m_B (k_B)^2 = 700(0.55)^2 = 212 \text{ kg-m}^2$$

We know that $\quad l_A.I_A = l_B.I_B \quad$ or $\quad l_A = \dfrac{l_B.I_B}{I_A} = \dfrac{l_B \times 212}{650} = 0.326\, l_B$

Also, $\quad l_A + l_B = l = 8.95 \text{ m} \quad$ or $\quad 0.326\, l_B + l_B = 8.95 \quad$ or $\quad l_B = 6.75 \text{ m}$

and $\quad l_A = 8.95 - 6.75 = 2.2 \text{ m}$

Hence the node lies at 2.2 m from flywheel A or 6.75 m from flywheel B on the equivalent shaft.

$\therefore$ Position of node on the original shaft from flywheel A

$$= l_1 + (l_A - l_1)\left(\frac{d_2}{d_1}\right)^4 = 0.6 + (2.2 - 0.6)\left(\frac{0.06}{0.095}\right)^4 = 0.855 \text{ m} \textbf{ Ans.}$$

Natural frequency of free torsional vibrations

We know that polar moment of inertia of the equivalent shaft,

$$J = \frac{\pi}{32}(d_1)^4 = \frac{\pi}{32}(0.095)^4 = 8 \times 10^{-6} \text{ m}^4$$

Natural frequency of free torsional vibrations,

$$f_n = f_{nA} \text{ or } f_{nB}$$
$$= \frac{1}{2\pi}\sqrt{\frac{C.J}{l_A.I_A}} = \frac{1}{2\pi}\sqrt{\frac{80 \times 10^9 \times 8 \times 10^{-6}}{2.2 \times 650}} = 3.37 \text{ Hz} \qquad \textbf{Ans.}$$

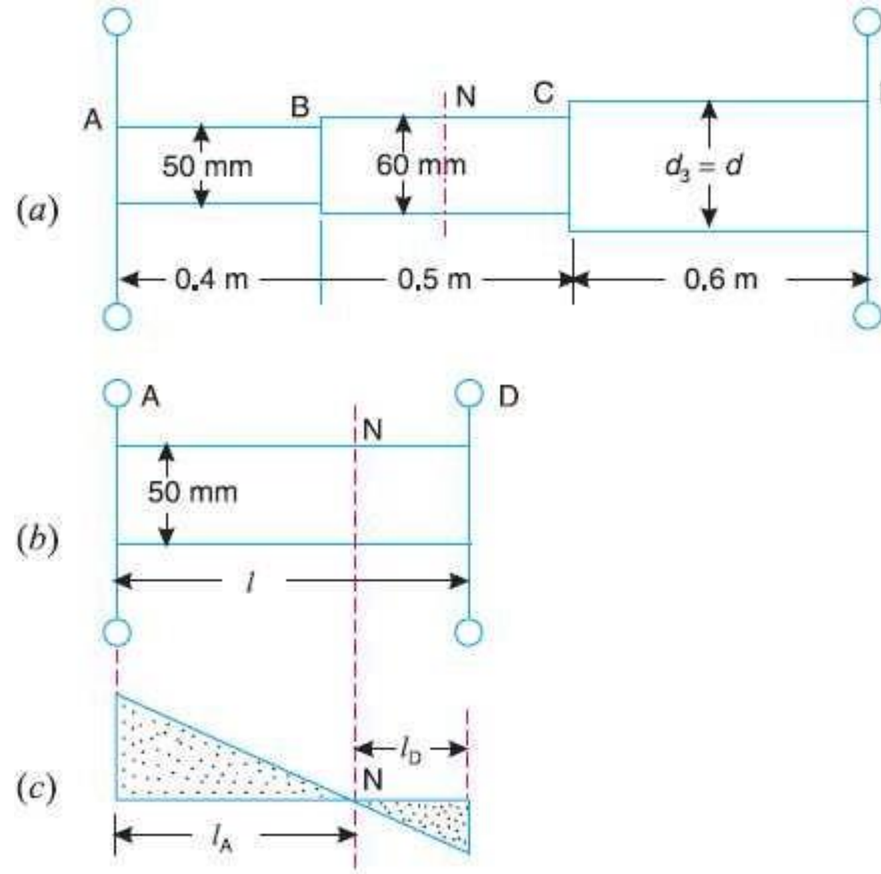
Example *A steel shaft ABCD 1.5 m long has flywheel at its ends A and D. The mass of the flywheel A is 600 kg and has a radius of gyration of 0.6 m. The mass of the flywheel D is 800 kg and has a radius of gyration of 0.9 m. The connecting shaft has a diameter of 50 mm for the portion AB which is 0.4 m long ; and has a diameter of 60 mm for the portion BC which is 0.5 m long ; and has a diameter of d mm for the portion CD which is 0.6 m long. Determine :*

1. the diameter 'd' of the portion CD so that the node of the torsional vibration of the system will be at the centre of the length BC ; and 2. the natural frequency of the torsional vibrations.

The modulus of rigidity for the shaft material is 80 GN/m².

Solution. Given : $L = 1.5$ m ; $m_A = 600$ kg ; $k_A = 0.6$ m ; $m_D = 800$ kg ; $k_D = 0.9$ m ; $d_1 = 50$ mm $= 0.05$ m ; $l_1 = 0.4$ m ; $d_2 = 60$ mm $= 0.06$ m ; $l_2 = 0.5$ m ; $d_3 = d$; $l_3 = 0.6$ m ; $C = 80$ GN/m² $= 80 \times 10^9$ N/m²

The actual shaft is shown is Fig. 24.10 (*a*). First of all, let us find the length of the equivalent shaft, assuming its diameter as $d_1 = 50$ mm, as shown in Fig. 24.10 (*b*).



We know that length of the equivalent shaft,

$$l \quad l_1 \quad l_2 \quad \frac{d_1}{d_2}^{4} \quad l_3 \quad \frac{d_1}{d_3}^{4} \quad 0.4 \quad 0.5 \quad \frac{0.05}{0.06}^{4} \quad 0.6 \quad \frac{0.05}{d}^{4}$$

. . . [Substituting $d_3 = d$]

$$0.4 \quad 0.24 \quad \frac{3.75 \quad 10^{6}}{d^4} \quad 0.64 \quad \frac{3.75 \quad 10^{6}}{d^4} \qquad \ldots (i)$$

1. *Diameter 'd' of the shaft CD*

Suppose the node of the equivalent shaft lies at N as shown in Fig. 24.10 (c).

Let l_A = Distance of the node from flywheel A, and l_D = Distance of the node from flywheel D.

We know that mass moment of inertia of flywheel A,

$$I_A \quad m_A (k_A)^2 \quad 600 (0.6)^2 = 216 \text{ kg-m}^2$$

and mass moment of inertia of flywheel D,

$$I_D \quad m_D (k_D)^2 \quad 800 (0.9)^2 = 648 \text{ kg-m}^2$$

We know that

$$l_A . I_A \quad l_D . I_D$$

or

$$l_D \quad \frac{l_A . I_A}{I_D} \quad \frac{l_A \quad 216}{648} \quad \frac{l_A}{3}$$

Since the node lies in the centre of the length BC in an original system, therefore its equivalent length from rotor A,

$$l_A \quad l_1 \quad \frac{l_2}{2} \; \frac{d_1}{d_2} \;^4 \quad 0.4 \quad \frac{0.5}{2} \; \frac{0.05}{0.06} \;^4 \quad 0.52 \text{ m}$$

$$l_D \quad \frac{l_A}{3} \quad \frac{0.52}{3} = 0.173 \text{ m}$$

We know that $l = l_A + l_D$

or

$$0.64 \quad \frac{3.75 \quad 10 \;^6}{d^4} \quad 0.52 \quad 0.173 \qquad \text{. . . [From equation } (i)]$$

$$\frac{3.75 \quad 10 \;^6}{d^4} \quad 0.52 \quad 0.173 \quad 0.64 \quad 0.053$$

$$d^4 \quad \frac{3.75 \quad 10 \;^6}{0.053} \quad 70.75 \quad 10 \;^6$$

or

$$d = 0.0917 \text{ m} = 91.7 \text{ mm} \quad \textbf{Ans.}$$

2. *Natural frequency of torsional vibrations*

We know that polar moment of inertia of the equivalent shaft,

$$J \quad \frac{}{2}(d_1)^4 \quad \frac{}{32}(0.05)^4 \quad 0.614 \quad 10 \;^6 \text{m}^4$$

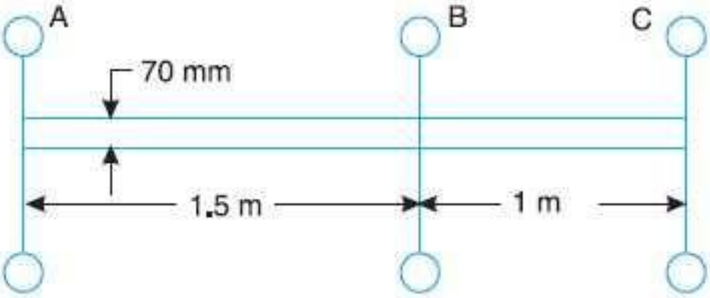
Natural frequency of torsional vibration,

$$f_n \quad f_{nA} \text{ or } f_{nD}$$

$$\frac{1}{2} \sqrt{\frac{C.J}{l_A . I_A}} \quad \frac{1}{2} \sqrt{\frac{80 \quad 10^9 \quad 0.614 \quad 10 \;^6}{0.52 \quad 216}} \text{ Hz} = 3.33 \text{ Hz} \quad \textbf{Ans.}$$

Example *A single cylinder oil engine drives directly a centrifugal pump. The rotating mass of the engine, flywheel and the pump with the shaft is equivalent to a three rotor system as shown in Fig. 24.11.*

The mass moment of inertia of the rotors A, B and C are 0.15, 0.3 and 0.09 kg-m^2. Find the natural frequency of the torsional vibration. The modulus of rigidity for the shaft material is 84 kN/mm^2.



Solution. Given : $I_A = 0.15$ kg-m^2 ; $I_B = 0.3$ kg-m^2 ; $I_C = 0.09$ kg-m^2 ; $d = 70$ mm $= 0.07$ m; $l_1 = 1.5$ m ; $l_2 = 1$ m ; $C = 84$ kN/mm^2 $= 84 \times 10^9$ N/m^2

We know that

$$l_A . I_A = l_C . I_C$$

$$\therefore \quad l_A = \frac{l_C . I_C}{I_A} = l_C \times \frac{0.09}{0.15} = 0.6\, l_C$$

Also

$$\frac{1}{l_C . I_C} = \frac{1}{I_B}\left[\frac{1}{l_1 - l_A} + \frac{1}{l_2 - l_C}\right]$$

$$\frac{1}{l_C \times 0.09} = \frac{1}{0.3}\left[\frac{1}{1.5 - l_A} + \frac{1}{1 - l_C}\right]$$

$$\frac{0.3}{l_C \times 0.09} = \frac{1}{1.5 - 0.6 l_C} + \frac{1}{1 - l_C} \qquad \text{. . . (Substituting } l_A = 0.6\, l_C)$$

$$\frac{10}{3 l_C} = \frac{1 - l_C + 1.5 - 0.6 l_C}{(1.5 - 0.6 l_C)(1 - l_C)} = \frac{2.5 - 1.6 l_C}{1.5 - 2.1 l_C + 0.6 (l_C)^2}$$

On cross-multiplying and re-arranging,

$$10.8\, (l_C)^2 - 28.5\, l_C + 15 = 0$$

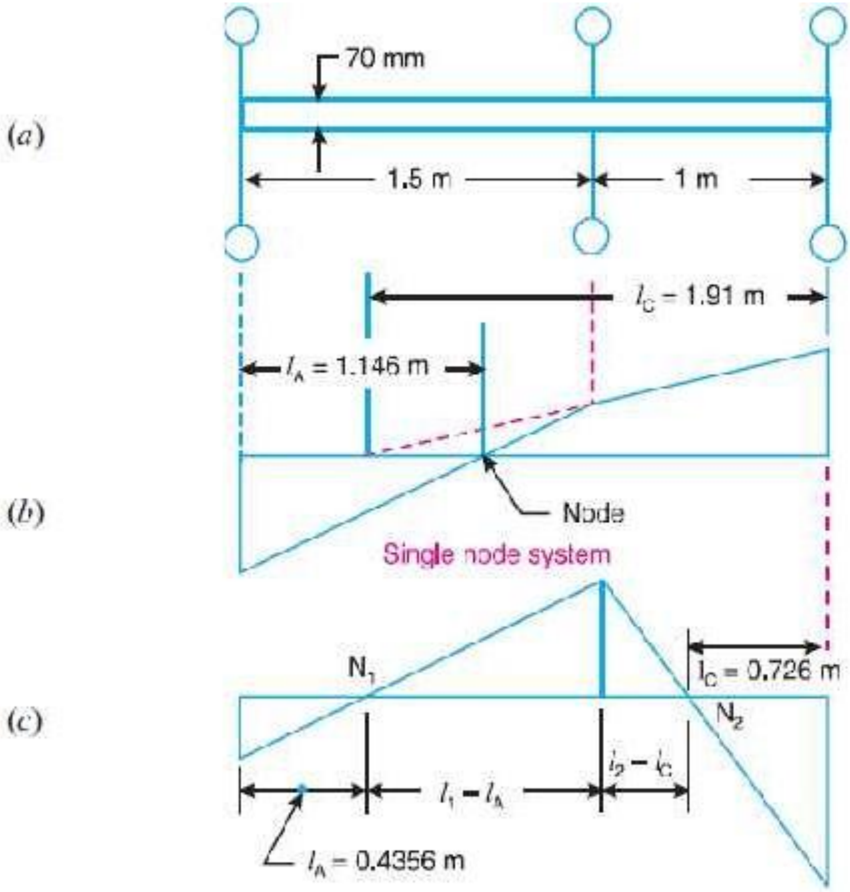
$$\therefore \quad l_C = \frac{28.5 \pm \sqrt{(28.5)^2 - 4 \times 10.8 \times 15}}{2 \times 10.8} = \frac{28.5 \pm 12.8}{21.6}$$

$$= 1.91 \text{ m} \quad \text{or} \quad 0.726 \text{ m}$$

and

$$l_A = 0.6\, l_C = 1.146 \text{ m} \quad \text{or} \quad 0.4356 \text{ m}$$

Here we see that when $l_C = 1.91$ m, then $l_A = 1.146$ m. This gives the position of single node for $l_A = 1.146$ m, as shown in Fig. 24.12 (*b*). The value of $l_C = 0.726$ m and corresponding value of $l_A = 0.4356$ m gives the position of two nodes, as shown in Fig. 24.12 (*c*).



We know that polar moment of inertia of the shaft,

$$J = \frac{\pi}{32} d^4 = \frac{\pi}{32}(0.07)^4 = 2.36 \times 10^{-6}\,\text{m}^4$$

Natural frequency of torsional vibration for a single node system,

$$f_{n1} = \frac{1}{2\pi}\sqrt{\frac{C.J}{l_A.I_A}} = \frac{1}{2\pi}\sqrt{\frac{84 \times 10^9 \times 2.36 \times 10^{-6}}{1.146 \times 0.15}}\ \text{Hz}$$

$= 171$ Hz **Ans.** . . . (Substituting $l_A = 1.146$ m)

Similarly, natural frequency of torsional vibration for a two node system,

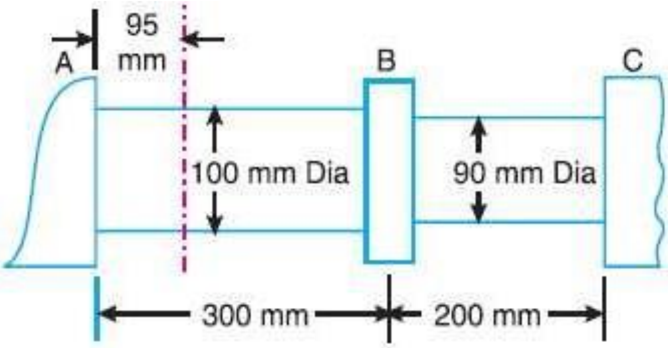
$$f_{n2} = \frac{1}{2\pi}\sqrt{\frac{C.J}{l_A.I_A}} = \frac{1}{2\pi}\sqrt{\frac{84 \times 10^9 \times 2.36 \times 10^{-6}}{0.4356 \times 0.15}}\ \text{Hz}$$

$= 277$ Hz **Ans.** . . . (Substituting $l_A = 0.4356$ m)

Example *A motor generator set, as shown in Fig. 24.13, consists of two armatures A and C connected with flywheel between them at B. The modulus of rigidity of the connecting shaft is 84 GN/m². The system can vibrate torsionally with one node at 95 mm from A, the flywheel being at antinode. Find : **1.** the position of another node ; **2.** the natural frequency of vibration; and **3.** the radius of gyration of the armature C.*

The other data are given below :

Particulars	*A*	*B*	*C*
Radius of gyration, mm	*300*	*375*	–
Mass, kg	*400*	*500*	*300*



Solution. Given : $C = 84$ GN/m² $= 84 \times 10^9$ N/m² ; $d_1 = 100$ mm $= 0.1$ m ; $d_2 = 90$ mm $= 0.09$ m ; $l_1 = 300$ mm $= 0.3$ m ; $l_2 = 200$ mm $= 0.2$ m ; $l_A = 95$ mm $= 0.095$ m ; $m_A = 400$ kg; $k_A = 300$ mm $= 0.3$ m ; $m_B = 500$ kg ; $k_B = 375$ mm $= 0.375$ m ; $m_C = 300$ kg

We know that mass moment of inertia of armature *A*,

$$I_A = m_A (k_A)^2 = 400 (0.3)^2 = 36 \text{ kg-m}^2$$

and mass moment of inertia of flywheel *B*,

$$I_B = m_B (k_B)^2 = 500 (0.375)^2 = 70.3 \text{ kg-m}^2$$

1. *Position of another node*

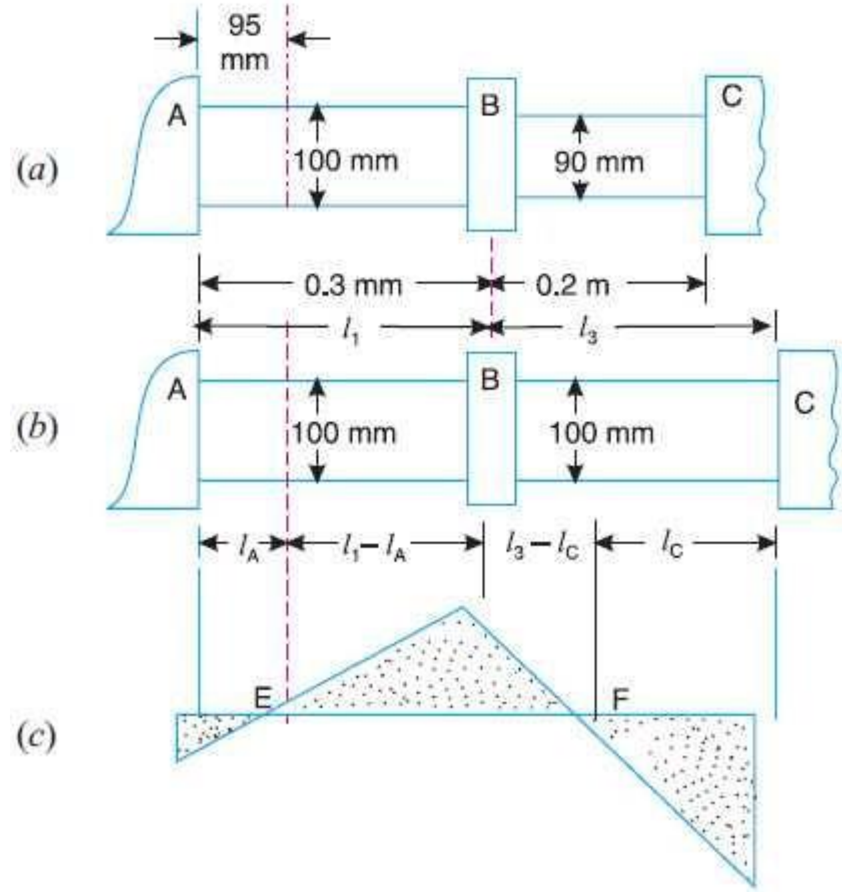
First of all, replace the original system, as shown in Fig. 24.14 (*a*), by an equivalent system as shown in Fig. 24.14 (*b*). It is assumed that the diameter of the equivalent shaft is $d_1 = 100$ mm $= 0.1$ m because the node lies in this portion. We know that the length of the equivalent shaft,

$$l = l_1 + l_2 \left(\frac{d_1}{d_2}\right)^4 = 0.3 + 0.2 \left(\frac{0.1}{0.09}\right)^4 = 0.605 \text{ m}$$

The first node lies at *E* at a distance 95 mm from rotor *A i.e.* $l_A = 95$ mm $= 0.095$ m, as shown in Fig. 24.14 (*c*).

Let l_C = Distance of node *F* in an equivalent system from rotor *C*, and

l_3 = Distance between flywheel *B* and armature *C* in an equivalent system $= l - l_1 = 0.605 - 0.3 = 0.305$ m



We know that

$$\frac{1}{I_B} \quad \frac{1}{l_1 \quad l_A} \quad \frac{1}{l_3 \quad l_C} \qquad \frac{1}{l_A . I_A}$$

$$\frac{1}{70.3} \quad \frac{1}{0.3 \quad 0.095} \quad \frac{1}{0.305 \quad l_C} \qquad \frac{1}{0.095 \quad 36}$$

$$\frac{1}{0.205} \quad \frac{1}{0.305 \quad l_C} \quad \frac{70.3}{0.095 \quad 36} \quad 20.56$$

$$\frac{1}{0.305 \quad l_C} \quad 20.56 \quad \frac{1}{0.205} \quad 15.68$$

$$0.305 - l_C = 1 / 15.68 = 0.064 \text{ or } l_C = 0.21 \text{ m}$$

Corresponding value of l_C in an original system from rotor C

$$0.21 \quad \frac{d_2}{d_1}^{\;4} \quad 0.21 \quad \frac{0.09}{0.1}^{\;4} = 0.13 \text{ m}$$ **Ans.**

2. ***Natural frequency of vibration***

We know that polar moment of inertia of the equivalent shaft,

$$J \quad \frac{}{32}(d_1)^4 \quad \frac{}{32}(0.1)^4 \quad 9.82 \quad 10 \ ^6\text{m}^4$$

Natural frequency of vibrations,

$$f_n \quad \frac{1}{2}\sqrt{\frac{C.J}{l_A \cdot I_A}} \quad \frac{1}{2}\sqrt{\frac{84 \quad 10^9 \quad 9.82 \quad 10 \ ^6}{0.095 \quad 36}} \quad \text{Hz}$$

$$= 78.1 \text{ Hz } \textbf{Ans.}$$

3. ***Radius of gyration of armature C***

Let k_C = Radius of gyration of armature C in metres, and

I_C = Mass moment of inertia of armature $C = m_C\,(k_C)^2$ in kg-m^2.

We know that $\quad l_A \, I_A \quad l_C \, I_C \quad l_C \, m_C \, (k_C)^2$

or $$(k_C)^2 \quad \frac{l_A \, I_A}{l_C \, m_C} \quad \frac{0.095 \quad 36}{0.21 \quad 300} \quad 0.0543 \, \text{m}^2$$

$$k_C = 0.233 \text{ m } \textbf{Ans.}$$

Example *A 4-cylinder engine and flywheel coupled to a propeller are approximated to a 3-rotor system in which the engine is equivalent to a rotor of moment of inertia 800 kg-m², the flywheel to a second rotor of 320 kg-m² and the propeller to a third rotor of 20 kg-m². The first and the second rotors being connected by 50 mm diameter and 2 metre long shaft and the second and the third rotors being connected by a 25 mm diameter and 2 metre long shaft.*

Neglecting the inertia of the shaft and taking its modulus of rigidity as 80 GN/m², determine: **1.** *Natural frequencies of torsional oscillations, and* **2.** *The positions of the nodes.*

Solution. Given : $I_A = 800$ kg-m^2 ; $I_B = 320$ kg-m^2 ; $I_C = 20$ kg-m^2 ; $d_1 = 50$ mm $= 0.05$ m; $l_1 = 2$ m ; $d_2 = 25$ mm $= 0.025$ m ; $l_2 = 2$ m ; $C = 80 \times 10^9$ N/m^2

1. ***Natural frequencies of torsional oscillations***

First of all, replace the original system, as shown in Fig. 24.15 (*a*), by an equivalent system as shown in Fig. 24.15 (*b*). It is assumed that the diameter of equivalent shaft is $d_1 = 50$ mm $= 0.05$m.

We know that length of equivalent shaft,

$$l \quad l_1 \quad l_2 \quad \frac{d_1}{d_2}^4 \quad 2 \quad 2 \quad \frac{0.05}{0.025}^4 \quad 34 \, \text{m}$$

Now let us find the position of nodes for the equivalent system.

Let l_A = Distance of node N_1 from rotor A, and

l_C = Distance of node N_2 from rotor C.

We know that $\quad l_A \, I_A \quad l_C \, I_C$

$$l_A \quad \frac{l_C \, I_C}{I_A} \quad \frac{l_C \quad 20}{800} \quad 0.025 \, l_C$$

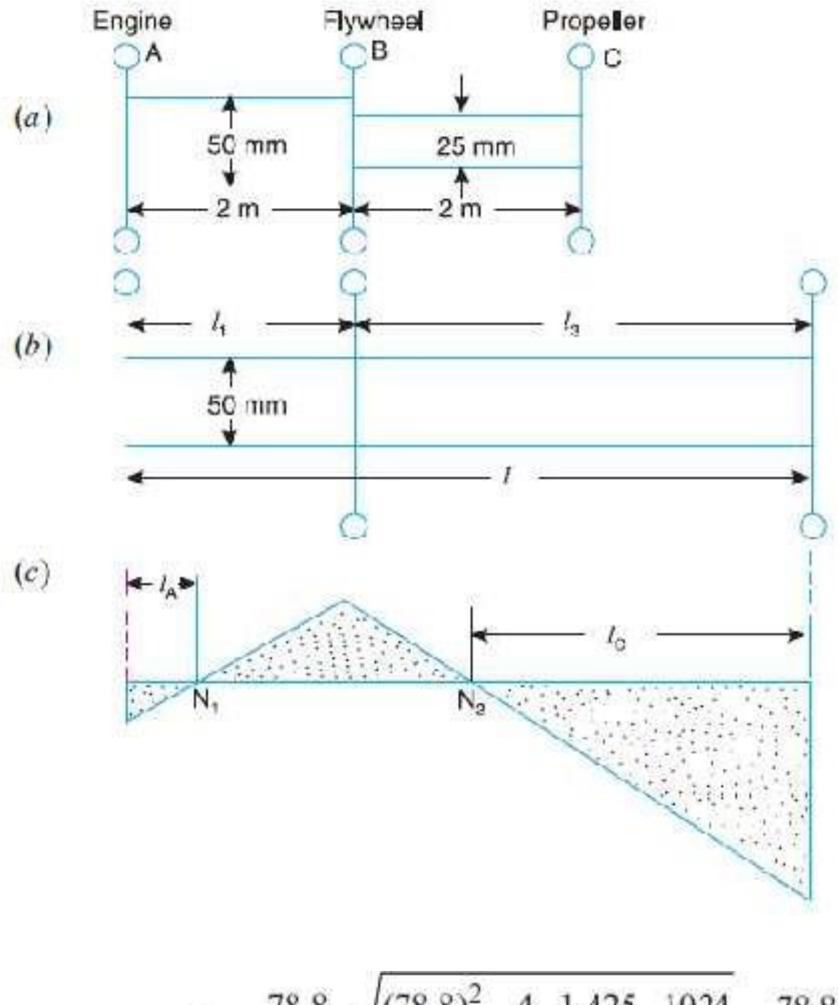
Also, $$\frac{1}{l_C \cdot I_C} \quad \frac{1}{I_B} \; \frac{1}{l_1 \quad l_A} \quad \frac{1}{l_3 \quad l_C}$$

$$\frac{1}{l_C \quad 20} \quad \frac{1}{320} \; \frac{1}{2 \quad 0.025 l_C} \quad \frac{1}{32 \quad l_C} \qquad \ldots (\qquad l_3 = l - l_1)$$

$$\frac{320}{l_C \quad 20} \quad \frac{1}{2 \quad 0.025 l_C} \quad \frac{1}{32 \quad l_C}$$

$$\frac{16}{l_C} \quad \frac{32 \quad l_C \quad 2 \quad 0.025 l_C}{(2 \quad 0.025 l_C)(32 \quad l_C)} \quad \frac{34 \quad 1.025 l_C}{64 \quad 2.8 l_C \quad 0.025 (l_C)^2}$$

or $\quad 1.425\,(l_C)^2 \quad 78.8\, l_C + 1024 - 0$



$$l_C \quad \frac{78.8 \quad \sqrt{(78.8)^2 \quad 4 \quad 1.425 \quad 1024}}{2 \quad 1.425} \quad \frac{78.8 \quad 19.3}{2.85}$$

$$= 34.42 \text{ m} \quad \text{or} \quad 20.88 \text{ m}$$

and $$l_A = 0.025\, l_C = 0.86 \text{ m} \quad \text{or} \quad 0.52 \text{ m}$$

We see that when $l_C = 34.42$ m, then $l_A = 0.86$ m. This gives the position of single node for $l_A = 0.86$ m. The value of $l_C = 20.88$ m and corresponding value of $l_A = 0.52$ m gives the position of two nodes, as shown in Fig. 24.15 (*c*).

We know that polar moment of inertia of the equivalent shaft,

$$J \quad \frac{\quad}{32}(d_1)^4 \quad \frac{\quad}{32}(0.05)^4 \quad 0.614 \quad 10^{\ 6}\,\mathrm{m}^4$$

Natural frequency of torsional vibrations for a single node system,

$$f_{n1} \quad \frac{1}{2}\sqrt{\frac{C.J}{l_A . I_A}} \quad \frac{1}{2}\sqrt{\frac{80 \quad 10^9 \quad 0.614 \quad 10^{\ 6}}{0.86 \quad 800}} \ \mathrm{Hz}$$

. . . (Substituting l_A = 0.86 m)

= 1.345 Hz **Ans.**

Similarly natural frequency of torsional vibrations for a two node system,

$$f_{n2} \quad \frac{1}{2}\sqrt{\frac{C.J}{l_A . I_A}} \quad \frac{1}{2}\sqrt{\frac{80 \quad 10^9 \quad 0.614 \quad 10^{\ 6}}{0.52 \quad 800}} \ \mathrm{Hz}$$

. . . (Substituting l_A = 0.52 m)

= 1.73 Hz **Ans.**

2. Position of the nodes

We have already calculated that for a two node system on an equivalent shaft, l_C = 20.88 m from the propeller.

Corresponding value of l_C in an original system from the propeller

$$20.88 \quad \frac{d_2}{d_1}^{\ 4} \quad 20.88 \quad \frac{0.025}{0.05}^{\ 4} \quad 1.3 \ \mathrm{m}$$

Therefore one node occurs at a distance of l_A – 0.52 m from the engine and the other node at a distance of l_C = 1.3 m from the propeller. **Ans.**

FREE TORSIONAL VIBRATIONS OF A GEARED SYSTEM

Let d_1 and d_2 = Diameter of the shafts C and D,

l_1 and l_2 = Length of the shafts C and D,

I_A and I_B = Mass moment of inertia of the rotors A and B,

$_A$ and $_B$ = Angular speed of the rotors A and B,

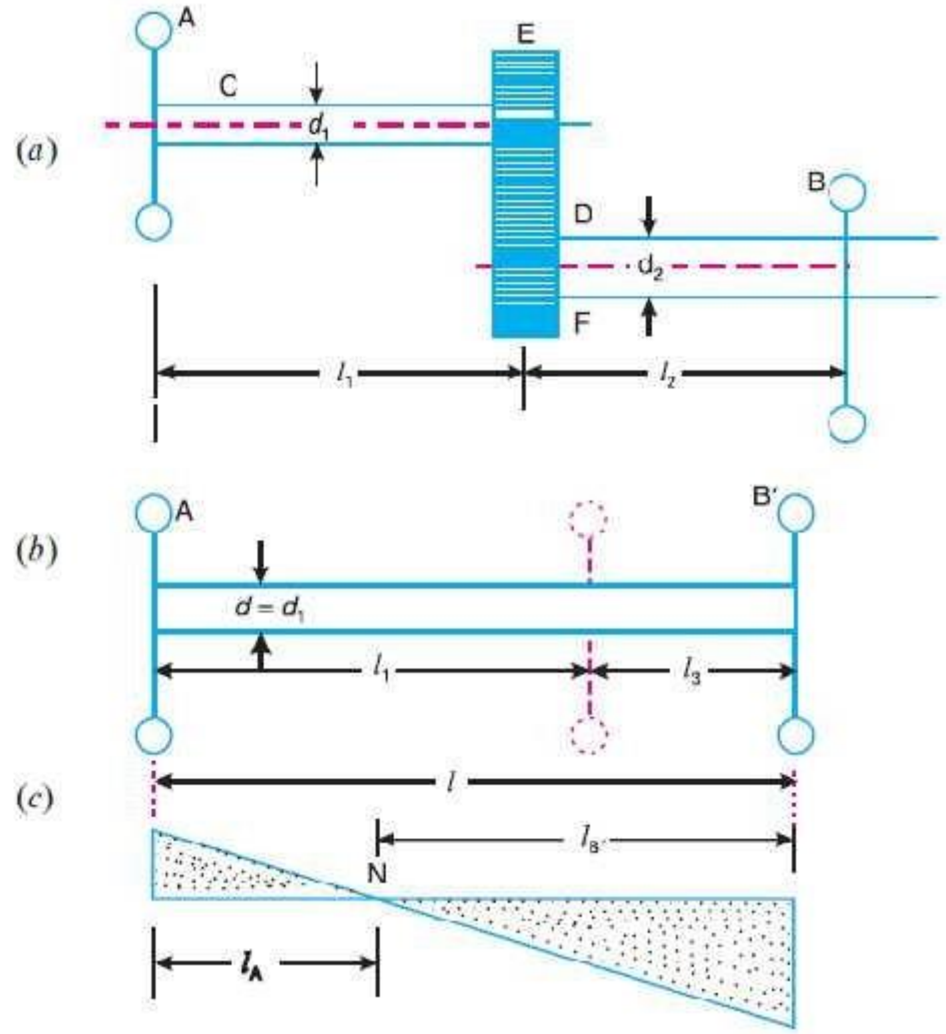
$$G \quad \text{Gear ratio} \quad \frac{\text{Speed of pinion } E}{\text{Speed of wheel } F} \quad \frac{\ _A}{\ _B}$$

. . . (Speeds of E and F will be same as that of rotors A and B)

d = Diameter of the equivalent shaft,

l = Length of the equivalent shaft, and

I_B = Mass moment of inertia of the equivalent rotor B .



$$\frac{T_3}{T_2} \quad \frac{2}{3} \quad \frac{1}{G}$$

where T_2 and T_3 = Torque on the sections l_2 and l_3, and

$_2$ and $_3$ = Angle of twist on sections l_2 and l_3.

Length of the equivalent shaft,

$$l = l_1 + \ l_3 = l_1 + G^2.l_2 \quad \frac{d_1}{d_2}^{\ 4}$$

NATURAL FREQUENCY OF TORSIONAL VIBRATION OF A GEARED SYSTEM:

Let the node of the equivalent system lies at N as shown in Fig. (c), then the natural frequency of torsional vibration of rotor A,

$$f_{nA} \quad \frac{1}{2}\sqrt{\frac{C.J}{l_A.I_A}}$$

and natural frequency of the torsional vibration of rotor B,

$$f_{nB} \quad \frac{1}{2}\sqrt{\frac{C.J}{l_B\ I_B}}$$

We know that $f_{nA} \quad f_{nB}$

$$\frac{1}{2}\sqrt{\frac{C.J}{l_A.I_A}} \quad \frac{1}{2}\sqrt{\frac{C.J}{l_B\ I_B}}$$

or

$$l_A.I_A \quad l_B\ I_B$$

$$l_A \quad l_B \quad l$$

PROBLEMS:

Example *A motor drives a centrifugal pump through gearing, the pump speed being one- third that of the motor. The shaft from the motor to the pinion is 60 mm diameter and 300 mm long. The moment of inertia of the motor is 400 kg-m^2. The impeller shaft is 100 mm diameter and 600 mm long. The moment of inertia of the impeller is 1500 kg-m^2. Neglecting inertia of the gears and the shaft, determine the frequency of torsional vibration of the system. The modulus of rigidity of the shaft material is 80 GN/m^2.*

Solution. Given : $G = N_A/N_B = 3$; $d_1 = 60$ mm $= 0.06$ m ; $l_1 = 300$ mm $= 0.3$ m ; $I_A = 400$ kg-m^2; $d^2 = 100$ mm $= 0.1$ m ; $l_2 = 600$ mm $= 0.6$ m ; $I_B = 1500$ kg-m^2 ; $C = 80$ GN/m$^2 = 80 \times 10^9$ N/m^2

The original and the equivalent system, neglecting the inertia of the gears, is shown in Fig. 24.17 (a) and (b) respectively. First of all, let us find the mass moment of inertia of the equivalent rotor B and the additional length of the equivalent shaft, assuming its diameter as $d_1 = 60$ mm.

We know that mass moment of the equivalent rotor B,

$$I_B \quad I_B/G^2 \quad 1500/3^2 \quad 166.7 \text{ kg–m}^2$$

and additional length of the equivalent shaft,

$$l_3 \quad G^2\, l_2 \ \frac{d_1}{d_2}^{4} \quad 3^2 \ \ 0.6 \ \frac{0.06}{0.1}^{4} \quad 0.7\text{m} \quad 700 \text{ mm}$$

Total length of the equivalent shaft,

$$l \quad l_1 \quad l_3 \quad 300 \quad 700 \quad 1000 \text{ mm} \quad 1 \text{ m}$$

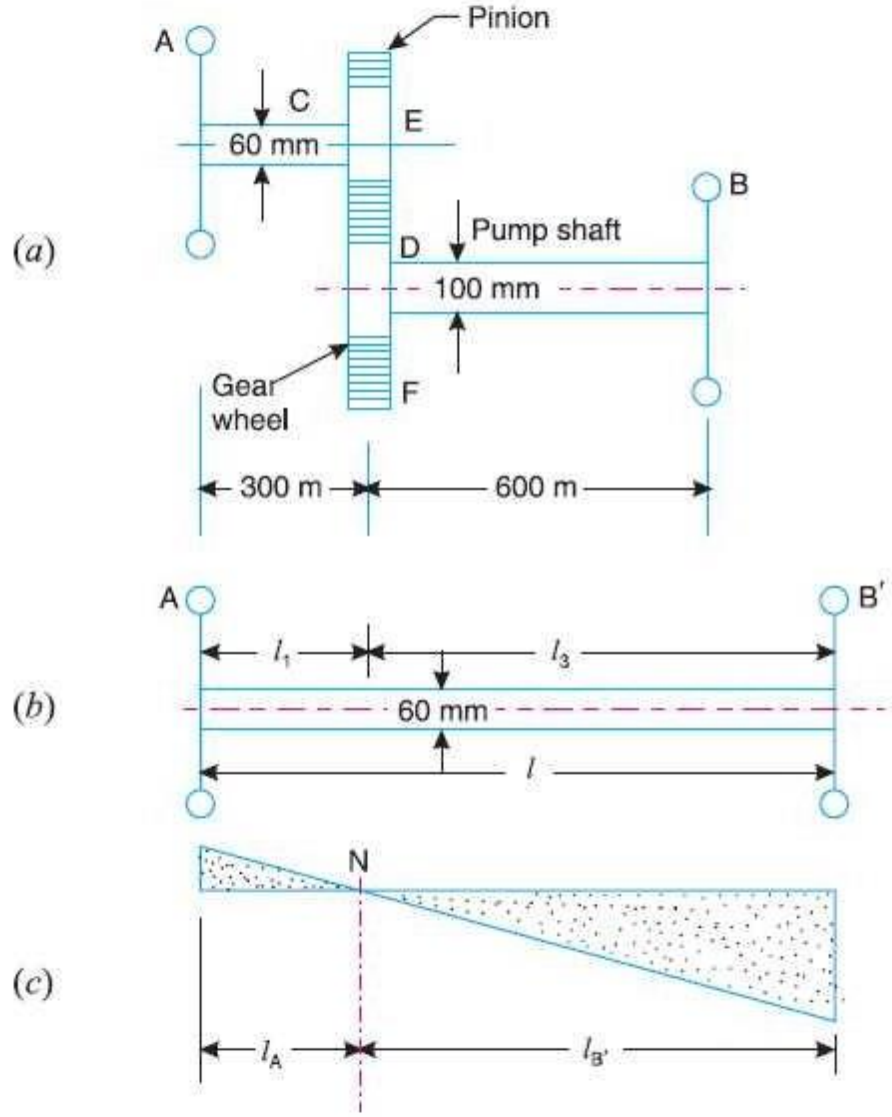
Let the node of the equivalent system lies at N, as shown in Fig. 24.17 (*c*). We know that

$$l_A . I_A \quad l_B . I_B \quad \text{or} \quad l_A \quad 400 \quad (1 \quad l_A)166.7 \; \ldots \; (\quad l_B \quad l \quad l_A)$$

$$l_A \quad 0.294 \text{ m} \quad 294 \text{ mm}$$

We know that polar moment of inertia of the equivalent shaft,

$$J \quad \frac{}{32}(d_1)^4 \quad \frac{}{32}(0.06)^4 \quad 1.27 \quad 10^{\;\;6} \text{m}^4$$



Frequency of torsional vibration,

$$f_n \quad \frac{1}{2}\sqrt{\frac{C.J}{l_A . I_A}} \quad \frac{1}{2}\sqrt{\frac{80 \quad 10^9 \quad 1.27 \quad 10^{\;\;6}}{0.294 \quad 400}} \text{ Hz}$$

$= 4.7$ Hz **Ans.**

Example *An electric motor is to drive a centrifuge, running at four times the motor speed through a spur gear and pinion. The steel shaft from the motor to the gear wheel is 54 mm diameter and L metre long ; the shaft from the pinion to the centrifuge is 45 mm diameter and 400 mm long. The masses and radii of gyration of motor and centrifuge are respectively 37.5 kg, 100 mm ; 30 kg and 140 mm.*

Neglecting the inertia effect of the gears, find the value of L if the gears are to be at the node for torsional oscillation of the system and hence determine the frequency of torsional oscillation. Assume modulus of rigidity for material of shaft as 84 GN/m².

Solution. Given : $G = N_A/N_B = 1/4 = 0.25$; $d_1 = 54$ mm $= 0.054$ m ; $l_1 = L$ m ; $d_2 = 45$ mm $= 0.045$ m ; $l_2 = 400$ mm $= 0.4$ m ; $m_A = 37.5$ kg ; $k_A = 100$ mm $= 0.1$ m ; $m_B = 30$ kg ; $k_B - 140$ mm $- 0.14$ m; $C - 84$ GN/m^2 $- 84 \times 10^9$ N/m^2

Value of L

We know that mass moment of inertia of the motor,

$$I_A \quad m_A (k_A)^2 \quad 37.5(0.1)^2 \quad 0.375 \text{ kg-m}^2$$

and mass moment of inertia of the centrifuge,

$$I_B \quad m_B (k_B)^2 \quad 30(0.14)^2 \quad 0.588 \text{ kg-m}^2$$

The original and the equivalent system, neglecting the inertia effect of the gears, is shown in Fig. 24.18 (*a*) and (*b*) respectively.

First of all, let us find the mass moment of inertia of the equivalent rotor B and the additional length of the equivalent shaft, keeping the diameter of the equivalent shaft as $d_1 = 54$ mm.

We know that mass moment of inertia of the equivalent rotor B ,

$$I_B \quad I_B / G^2 \quad 0.588/(0.25)^2 \quad 9.4 \text{ kg-m}^2$$

and additional length of equivalent shaft,

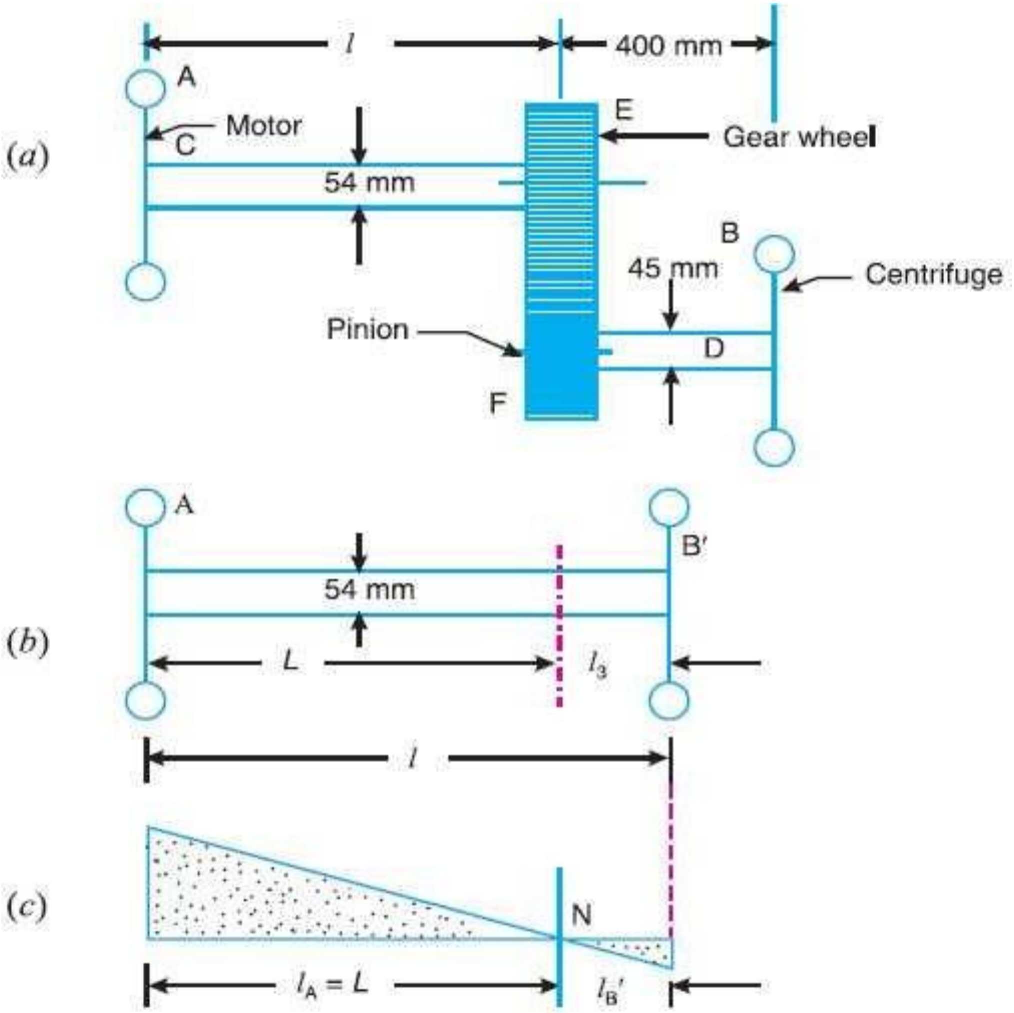
$$l_3 \quad G^2 l_2 \ \frac{d_1}{d_2}^4 \quad (0.25)^2 0.4 \ \frac{0.054}{0.045}^4 \quad 0.0518 \text{ m}$$

Since the node N for torsional oscillation of the system lies at the gears, as shown in Fig. 24.18 (*c*), therefore

$$l_A \quad L, \quad \text{and} \quad l_B \quad l_3 \quad 0.0518 \text{ m}$$

We know that $\quad l_A . I_A \quad l_B . I_B$

$$L \times 0.375 = 0.0518 \times 9.4 = 0.487 \quad \text{or } L = 1.3 \text{ m} \quad \textbf{Ans.}$$



Frequency of torsional oscillations

We know that polar moment of inertia of the equivalent shaft,

$$J \quad \frac{}{32}(d_1)^4 \quad \frac{}{32}(0.054)^4 \quad 0.835 \quad 10^{\;6}\,\mathrm{m}^4$$

Frequency of torsional oscillations,

$$f_n \quad \frac{1}{2}\sqrt{\frac{C.J}{l_A.I_A}} \quad \frac{1}{2}\sqrt{\frac{84 \quad 10^9 \quad 0.835 \quad 10^{\;6}}{1.3 \quad 0.375}} = 60.4 \text{ Hz } \textbf{Ans.}$$

Example *Determine the natural frequencies of torsional oscillation for the following system. The system is a reciprocating I.C. engine coupled to a centrifugal pump through a pair of gears. The shaft from the flywheel of the engine to the gear wheel is of 60 mm diameter and 950 mm length. The shaft from the pinion to the pump is of 40 mm diameter and 300 mm length. The engine speed is* $\frac{1}{4}$ *th of the pump speed.*

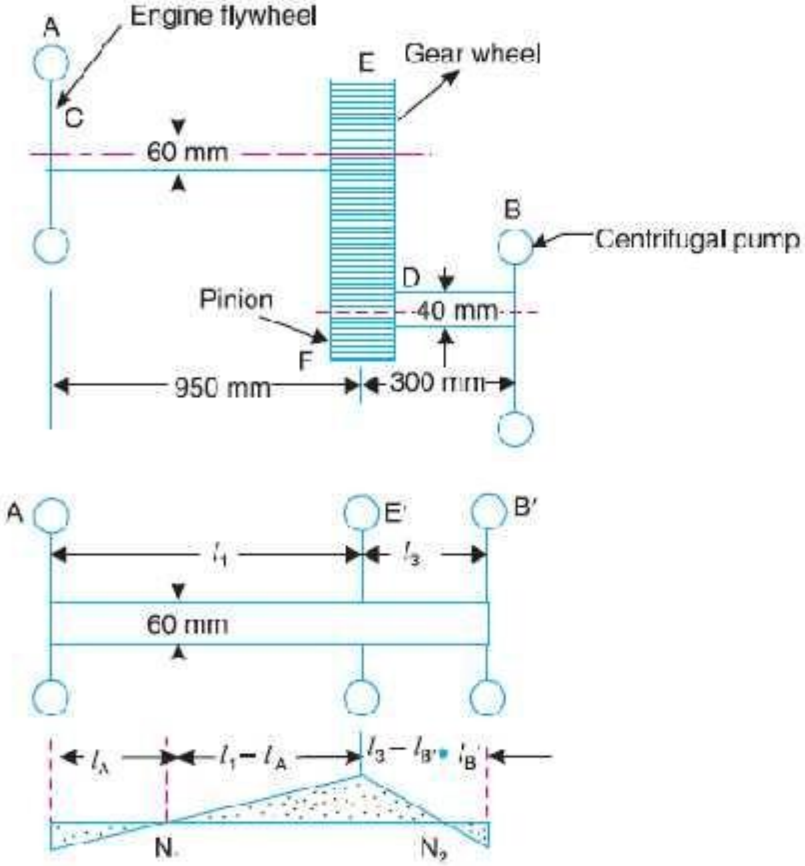
Moment of inertia of the flywheel = 800 kg-m²
Moment of inertia of the gear wheel = 15 kg-m²
Moment of inertia of the pinion = 4 kg-m²
Moment of inertia of the pump = 17 kg-m²
Modulus of rigidity for shaft material is 84 GN/m².

Solution. Given : $d_1 = 60$ mm $= 0.06$ m ; $l_1 = 950$ mm $= 0.95$ m ; $d_2 = 40$ mm $= 0.04$ m ; $l_2 = 300$ mm $= 0.3$ m ; $G = 1/4 = 0.25$; $I_A = 800$ kg-m^2 ; $I_E = 15$ kg-m^2 ; $I_F = 4$ kg-m^2 ; $I_B = 17$ kg-m^2; $C = 84$ GN/m$^2 = 84 \times 10^9$ N/m^2

The original and the equivalent system is shown in Fig. 24.19 (*a*) and (*b*) respectively. First of all, let us find the mass moment of inertia of the equivalent gearing E , the equivalent pump B and the additional length of the equivalent shaft keeping its diameter as $d_1 = 60$ mm.

We know that mass moment of inertia of the equivalent gearing E ,

$$I_E \quad I_E \quad I_F / G^2 \quad 15 \quad 4/(0.25)^2 \quad 79\ \text{kg-m}^2$$



All dimensions in mm.

Fig. 24.19

Mass moment of inertia of the equivalent pump B ,

$$I_B \quad I_B / G^2 \quad 17/(0.25)^2 \quad 272\ \text{kg-m}^2$$

and additional length of the equivalent shaft,

$$l_3 \quad G^2 . l_2 \quad \frac{d_1}{d_2}^4 \quad (0.25)^2 0.3 \quad \frac{0.06}{0.04}^4 \quad 0.095\,\text{m}$$

The original system is thus reduced to a three rotor system, as shown in 24.19 (*b*). Let us find the position of nodes for the equivalent system.

Let $\quad l_A$ = Distance of node N_1 from rotor A, and

l_B = Distance of node N_2 from rotor B .

We know that $\quad l_A . I_A \quad l_B . I_B \quad$ or $\quad l_A \quad l_B \; \frac{I_B}{I_A} \quad l_B \; \frac{272}{800} \quad 0.34 l_B$

Also $\quad \frac{1}{l_B . I_B} \quad \frac{1}{I_E} \; \frac{1}{l_1 \quad l_A} \; \frac{1}{l_3 \quad l_B}$

$$\frac{1}{l_B \quad 272} \quad \frac{1}{79} \; \frac{1}{0.95 \quad 0.34 l_B} \quad \frac{1}{0.095 \quad l_B}$$

$$\frac{79}{l_B \quad 272} \quad \frac{(0.095 \quad l_B \quad 0.95 \quad 0.34 l_B)}{(0.95 \quad 0.34 l_B)(0.095 \quad l_B)}$$

$$\frac{0.29}{l_B} \quad \frac{1.045 \quad 1.34 l_B}{0.09 \quad 0.98 l_B \quad 0.34 (l_B)^2}$$

$$0.026 \quad 0.28 l_B \quad 0.1 (l_B)^2 \quad 1.045 l_B \quad 1.34 (l_B)^2$$

$$1.44 (l_B)^2 \quad 1.325 l_B \quad 0.026 \quad 0$$

$$l_B \quad \frac{1.325 \quad \sqrt{(1.325)^2 \quad 4 \quad 1.44 \quad 0.026}}{2 \quad 1.44} \quad \frac{1.325 \quad 1.267}{2.88}$$

$$= 0.9 \text{ m or } 0.02 \text{ m}$$

and $\quad l_A \quad 0.34 l_B \quad 0.306 \text{ m} \;$ or $\; 0.0068 \text{ m}$

We see that when $l_B = 0.9$ m, then $l_A = 0.306$ m. This gives the position of single node for $l_A = 0.306$ m or 306 mm. When $l_B = 0.02$ m, the corresponding value of $l_A = 0.0068$ m or 6.8 mm. This gives the position of two nodes as shown in Fig. 24.19 (*c*).

We know that polar moment of inertia of the equivalent shaft,

$$J \quad \frac{}{32} (d_1)^4 \quad \frac{}{32} (0.06)^4 \quad 1.27 \quad 10 \; ^6 \text{ m}^4$$

Natural frequency of torsional oscillations for a single node system,

$$f_{n1} \quad \frac{1}{2} \sqrt{\frac{C.J}{l_A . I_A}} \quad \frac{1}{2} \sqrt{\frac{84 \quad 10^9 \quad 1.27 \quad 10 \; ^6}{0.306 \quad 800}} \text{ Hz}$$

$= 3.32$ Hz **Ans.** $\qquad$ . . . (Substituting $l_A = 0.306$ m)

Similarly natural frequency of torsional oscillations for a two node system,

$$f_{n2} \quad \frac{1}{2} \sqrt{\frac{C.J}{l_A . I_A}} \quad \frac{1}{2} \sqrt{\frac{84 \quad 10^9 \quad 1.27 \quad 10 \; ^6}{0.0068 \quad 800}} \text{ Hz}$$

$= 22.3$ Hz **Ans.** $\qquad$ . . . (Substituting $l_A = 0.0068$ m)